INCH-POUND

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SUPERSEDING
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15 November 1979

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, POSITIVE, VOLTAGE REGULATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for fixed output, monolithic silicon, positive voltage regulators. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete Part or Identifying Number (PIN) (see 6.6).

1.2 Classification.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Circuit	Case outline letter
01	Voltage regulator, +5 volts at 1.5 amperes (maximum)	X or Y (see table I)
02	Voltage regulator, +5 volts at 0.5 ampere	X
03	Voltage regulator, +12 volts at 0.5 ampere	X
04	Voltage regulator, +15 volts at 0.5 ampere	X
05	Voltage regulator, +24 volts at 0.5 ampere	X
06	Voltage regulator, +5 volts at 1.0 ampere	Y
07	Voltage regulator, +12 volts at 1.0 ampere	Y
08	Voltage regulator, +15 volts at 1.0 ampere	Y
09	Voltage regulator, +24 volts at 1.0 ampere	Y

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Case outline letter	Case outline
X	See figure 1
Y	See figure 2

1.3 Absolute maximum rating.

Input voltage:	
Device types 01, 02, 03, 04, 06, 07, and 08	35 V dc
Device types 05 and 09	40 V dc
Lead temperature (soldering, 60 seconds)	+300°C
Storage temperature	-65°C to +150°C
Junction temperature (T_J)	+150°C 1/

1/ The device is protected by a thermal shutdown circuit which is designed to turn off the output transistor whenever the device junction temperature is in excess of +150°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, AFSC, RADC/RBE-2, Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Input voltage range:	
Device type 01	+7.0 V dc to +25 V dc
Device types 02 and 06	+8.0 V dc to +25 V dc
Device types 03 and 07	+15 V dc to +32 V dc
Device types 04 and 08	+18.5 V dc to +35 V dc
Device types 05 and 09	+28 V dc to +38 V dc
Ambient operating temperature range (T_A)	-55°C to +125°C

1.5 Thermal characteristics:

$T_A = T_S$	Case	Maximum θ_{JA}	Maximum P_D without heat sink	Maximum θ_{JC}	Maximum P_D with heat sink	Maximum θ_{CS} 2/
+125°C 3/	X	140°C/W	0.18 W	40°C/W	0.5 W	10°C/W
	Y	35°C/W	0.71 W	4°C/W	5.6 W	0.5°C/W
+25°C	X	140°C/W	0.89 W	40°C/W	2.50 W	10°C/W
	Y	35°C/W	3.60 W	4°C/W	28.00 W	0.5°C/W
-55°C	X	140°C/W	1.50 W	40°C/W	4.00 W	10°C/W
	Y	35°C/W	5.80 W	4°C/W	45.00 W	0.5°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

2/ This value represents the maximum allowable thermal impedance of a heat sink to remain within the above thermal ratings.
3/ Based on T_J = +150°C and specified values of θ_{JA} and θ_{JC}.

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3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified in 1.2.3 herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Block diagram and terminal connections. The block diagram and terminal connections shall be as specified on figures 3 and 4.

3.2.2 Functional schematic circuits. The functional schematic circuits shall be as specified on figure 5. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.3 Case outlines. Case outlines shall be as specified in 1.2.3 and figures 1 and 2.

3.2.4 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4) and as follows.

3.3.1 Lead material.

Type X:

Nickel	50-53 percent
Manganese	0.60 percent, maximum
Silicon	0.30 percent, maximum
Carbon	0.10 percent, maximum
Chromium	0.25 percent, maximum
Cobalt	0.50 percent, maximum
Phosporous	0.025 percent, maximum
Sulfur	0.025 percent, maximum
Aluminum	0.10 percent, maximum
Iron	Remainder

Type Y:

Copper core	24.75 percent, maximum
Clad with Alloy 52	75.25 percent, maximum

3.3.2 Lead finish. Lead finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.4.1 Stability. If the device is located an appreciable distance from the power supply filter, a large (1.0 µF) bypass capacitor should be connected as close to the device V_{CC} input as possible to suppress oscillation. A (0.1 µF) bypass capacitor is recommended on the device output. Output current oscillations are likely to occur in the current limit operating mode with high input voltages (V_{IN} > 18 V dc).

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TABLE I. Electrical performance characteristics for device type 01 (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 14, unless otherwise specified)				Limits		Unit
		Input voltage	Load current		Other	Min	Max	
			I_L for case X	I_L for case Y				
Output voltage	V_{OUT}	V_{IN} = 7 V	-5 mA	-5 mA		4.70	5.40	V
		V_{IN} = 7 V, 8 V 1/	-0.5 A	-1.5 A		4.70	5.40	V
		V_{IN} = 9 V	-0.5 A			4.70	5.40	V
		V_{IN} = 18 V		-1.5 A		4.70	5.40	V
		V_{IN} = 25 V	-5 mA	-5 mA		4.70	5.40	V
		V_{IN} = 25 V	-100 mA	-1.0 A		4.70	5.40	V
Line regulation	V_{RLINE}	7 V $\leq V_{IN} \leq$ 25 V	-5 mA		T_A = +25°C	-50	+50	mV
					-55°C $\leq T_A \leq$ +125°C	-100	+100	mV
Load regulation	V_{RLOAD}	V_{IN} = 10 V	$I_{MAX} \leq I_L \leq$ -5 mA		-55°C $\leq T_A \leq$ +125°C	-100	+100	mV
			I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
Standby current drain	I_{SCD}	7 V $\leq V_{IN} \leq$ 25 V	-5 mA			-10.0	+0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	7 V $\leq V_{IN} \leq$ 25 V	-5 mA			-0.5	+0.5	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 01 (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 14, unless otherwise specified)				Limits		Unit
		Input voltage	Load current		Other	Min	Max	
			I_L for case X	I_L for case Y				
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 10 V	$I_{MAX} \leq I_L \leq$ -5 mA			-0.8	+0.8	mA
			I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
Output short 2/ circuit current	I_{OS}	V_{IN} = 35 V			Case X	-2.0	0.01	A
					Case Y	-2.8	0.01	A
Minimum start-up input voltage	V_{START}		Case X (R_L = 25Ω ±5%)		See figure 15		9.0	V
			Case Y (R_L = 5Ω ±5%)					
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 10 V e_i = 1 V rms at f = 2,400 Hz	-125 mA	-350 mA	T_A = +25°C (See figure 10)	60		dB
Output noise voltage 3/	N_O	V_{IN} = 10 V	-50 mA	-100 mA	T_A = +25°C (See figure 11)		125	μV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 10 V (V_{pulse} = 3.0 V)	-5 mA		T_A = +25°C (See figure 12)		15	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 10 V	-50 mA ΔI_L = -200 mA	-100 mA ΔI_L = -400 mA	T_A = +25°C (See figure 13)		2.0	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 7 V	-5 mA		T_A = +125°C and -55°C	-2.0	+2.0	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 02 4/ (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 8 V	I_L = -5 mA, -0.5 A		4.75	5.25	V
		V_{IN} = 20 V	I_L = -5 mA, -0.5 A		4.75	5.25	V
		V_{IN} = 35 V	I_L = -5 mA, -50 mA		4.75	5.25	V
		V_{IN} = 10 V	I_L = -5 mA	T_A = +150°C	4.70	5.30	V
Line regulation	V_{RLINE}	8 V ≤ V_{IN} ≤ 35 V	I_L = -50 mA		-150	+150	mV
		8 V ≤ V_{IN} ≤ 25 V	I_L = -350 mA		-50	+50	mV
Load regulation	V_{RLOAD}	V_{IN} = 10 V	-500 mA ≤ I_L ≤ -5 mA		-100	+100	mV
		V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA		-150	+150	mV
Standby current drain	I_{SCD}	V_{IN} = 10 V	I_L = -5 mA		-7.0	-0.5	mA
		V_{IN} = 35 V	I_L = -5 mA		-8.0	-0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	8 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 10 V	-500 mA ≤ I_L ≤ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 10 V			-2.00	-0.01	A
		V_{IN} = 25 V			-1.50	-0.01	A
		V_{IN} = 35 V			-1.00	-0.01	A
Overload current 6/	I_{OL}	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		See figure 9	-2.0	-0.5	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 02 4/ (see 3.4), unless otherwise specified - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 10 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	See figure 10 T_A = +25°C	60		dB
Output noise voltage 3/	N_O	V_{IN} = 10 V	I_L = -50 mA	See figure 11 T_A = +25°C		125	µV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 10 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 10 V	I_L = -50 mA ΔI_L = -200 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 10 V	I_L = -5 mA	T_A = +125°C and -55°C	-2.0	+2.0	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 03 4/ (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 15 V	I_L = -5 mA, -0.5 A		11.40	12.60	V
		V_{IN} = 27 V	I_L = -5 mA, -0.5 A		11.40	12.60	V
		V_{IN} = 35 V	I_L = -5 mA, -50 mA		11.40	12.60	V
		V_{IN} = 17 V	I_L = -5 mA	T_A = +150°C	11.28	12.72	V
Line regulation	V_{RLINE}	15 V ≤ V_{IN} ≤ 35 V	I_L = -50 mA		-360	+360	mV
		15 V ≤ V_{IN} ≤ 32 V	I_L = -350 mA		-120	+120	mV
Load regulation	V_{RLOAD}	V_{IN} = 17 V	-500 mA ≤ I_L ≤ -5 mA		-240	+240	mV
		V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA		-360	+360	mV
Standby current drain	I_{SCD}	V_{IN} = 17 V	I_L = -5 mA		-7.0	-0.5	mA
		V_{IN} = 35 V	I_L = -5 mA		-8.0	-0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 17 V	-500 mA ≤ I_L ≤ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 17 V			-1.75	-0.01	A
		V_{IN} = 32 V			-1.25	-0.01	A
		V_{IN} = 35 V			-1.00	-0.01	A
Overload current 6/	I_{OL}	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		See figure 9	-2.0	-0.5	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 03 4/ (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 17 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	See figure 10 T_A = +25°C	55		dB
Output noise voltage 3/	N_O	V_{IN} = 17 V	I_L = -50 mA	See figure 11 T_A = +25°C		250	µV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 17 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 17 V	I_L = -50 mA ΔI_L = -200 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 17 V	I_L = -5 mA	T_A = +125°C and -55°C	-3.0	+3.0	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 04 4/ (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 18.5 V	I_L = -5 mA, -0.5 A		14.25	15.75	V
		V_{IN} = 30 V	I_L = -5 mA, -0.5 A		14.25	15.75	V
		V_{IN} = 35 V	I_L = -5 mA, -50 mA		14.25	15.75	V
		V_{IN} = 20 V	I_L = -5 mA	T_A = +150°C	14.10	15.90	V
Line regulation	V_{RLINE}	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -350 mA		-150	+150	mV
Load regulation	V_{RLOAD}	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA		-300	+300	mV
		V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA		-450	+450	mV
Standby current drain	I_{SCD}	V_{IN} = 20 V	I_L = -5 mA		-7.0	-0.5	mA
		V_{IN} = 35 V	I_L = -5 mA		-8.0	-0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 20 V			-1.75	-0.01	A
		V_{IN} = 35 V			-1.00	-0.01	A
Overload current 6/	I_{OL}	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		See figure 9	-2.0	-0.5	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 04 4/ (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 20 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	See figure 10 T_A = +25°C	53		dB
Output noise voltage 3/	N_O	V_{IN} = 20 V	I_L = -50 mA	See figure 11 T_A = +25°C		300	µV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 20 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 20 V	I_L = -50 mA ΔI_L = -200 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 20 V	I_L = -5 mA	T_A = +125°C and -55°C	-3.75	+3.75	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 05 4/ (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 28 V	I_L = -5 mA, -0.5 A		22.80	25.20	V
		V_{IN} = 38 V	I_L = -5 mA, -0.5 A		22.80	25.20	V
		V_{IN} = 40 V	I_L = -5 mA, -50 mA		22.80	25.20	V
		V_{IN} = 30 V	I_L = -5 mA	T_A = +150°C	22.56	25.44	V
Line regulation	V_{RLINE}	28 V ≤ V_{IN} ≤ 40 V	I_L = -50 mA		-720	+720	mV
		28 V ≤ V_{IN} ≤ 38 V	I_L = -350 mA		-240	+240	mV
Load regulation	V_{RLOAD}	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA		-480	+480	mV
		V_{IN} = 40 V	-50 mA ≤ I_L ≤ -5 mA		-720	+720	mV
Standby current drain	I_{SCD}	V_{IN} = 30 V	I_L = -5 mA		-7.0	-0.5	mA
		V_{IN} = 40 V	I_L = -5 mA		-8.0	-0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	28 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 30 V			-1.25	-0.01	A
		V_{IN} = 38 V			-1.00	-0.01	A
		V_{IN} = 40 V			-1.00	-0.01	A
Overload current 6/	I_{OL}	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		See figure 9	-2.0	-0.5	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 05 4/ (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 30 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	See figure 10 T_A = +25°C	50		dB
Output noise voltage 3/	N_O	V_{IN} = 30 V	I_L = -50 mA	See figure 11 T_A = +25°C		500	µV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 30 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 30 V	I_L = -50 mA ΔI_L = -200 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 30 V	I_L = -5 mA	T_A = +125°C and -55°C	-6.0	+6.0	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 06 4/ (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 8 V	I_L = -5 mA, -1.0 A		4.75	5.25	V
		V_{IN} = 20 V	I_L = -5 mA, -1.0 A		4.75	5.25	V
		V_{IN} = 35 V	I_L = -5 mA, -0.1 A		4.75	5.25	V
		V_{IN} = 10 V	I_L = -5 mA	T_A = +150°C	4.70	5.30	V
Line regulation	V_{RLINE}	8 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A		-150	+150	mV
		8 V ≤ V_{IN} ≤ 25 V	I_L = -0.5 A		-50	+50	mV
Load regulation	V_{RLOAD}	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA		-100	+100	mV
		V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA		-150	+150	mV
Standby current drain	I_{SCD}	V_{IN} = 10 V	I_L = -5 mA		-7.0	-0.5	mA
		V_{IN} = 35 V	I_L = -5 mA		-8.0	-0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	8 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 10 V			-4.00	-0.02	A
		V_{IN} = 25 V			-3.00	-0.02	A
		V_{IN} = 35 V			-2.00	-0.02	A
Overload current 6/	I_{OL}	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		See figure 9	-4.0	-1.0	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 06 4/ (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 10 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	See figure 10 T_A = +25°C	60		dB
Output noise voltage 3/	N_O	V_{IN} = 10 V	I_L = -0.1 A	See figure 11 T_A = +25°C		125	µV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 10 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 10 V	I_L = -100 mA ΔI_L = -400 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 10 V	I_L = -5 mA	T_A = +125°C and -55°C	-2.0	+2.0	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 07 4/ (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 15 V	I_L = -5 mA, -1.0 A		11.40	12.60	V
		V_{IN} = 27 V	I_L = -5 mA, -1.0 A		11.40	12.60	V
		V_{IN} = 35 V	I_L = -5 mA, -0.1 A		11.40	12.60	V
		V_{IN} = 17 V	I_L = -5 mA	T_A = +150°C	11.28	12.72	V
Line regulation	V_{RLINE}	15 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A		-360	+360	mV
		15 V ≤ V_{IN} ≤ 32 V	I_L = -0.5 A		-120	+120	mV
Load regulation	V_{RLOAD}	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA		-240	+240	mV
		V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA		-360	+360	mV
Standby current drain	I_{SCD}	V_{IN} = 17 V	I_L = -5 mA		-7.0	-0.5	mA
		V_{IN} = 35 V	I_L = -5 mA		-8.0	-0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 17 V			-3.50	-0.02	A
		V_{IN} = 32 V			-2.50	-0.02	A
		V_{IN} = 35 V			-2.00	-0.02	A
Overload current 6/	I_{OL}	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		See figure 9	-4.0	-1.0	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 07 4/ (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 17 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	See figure 10 T_A = +25°C	55		dB
Output noise voltage 3/	N_o	V_{IN} = 17 V	I_L = -0.1 A	See figure 11 T_A = +25°C		250	µV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 17 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 17 V	I_L = -100 mA ΔI_L = -400 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 17 V	I_L = -5 mA	T_A = +125°C and -55°C	-3.0	+3.0	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 08 4/ (see 3.4), unless otherwise specified).

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 18.5 V V_{IN} = 30 V V_{IN} = 35 V	I_L = -5 mA, -1.0 A I_L = -5 mA, -1.0 A I_L = -5 mA, -0.1 A		14.25 14.25 14.25	15.75 15.75 15.75	V V V
		V_{IN} = 20 V	I_L = -5 mA	T_A = +150°C	14.10	15.90	V
Line regulation	V_{RLINE}	18.5 V $\leq V_{IN} \leq$ 35 V	I_L = -0.5 A		-150	+150	mV
Load regulation	V_{RLOAD}	V_{IN} = 20 V V_{IN} = 35 V	-1.0 A $\leq I_L \leq$ -5 mA -0.1 A $\leq I_L \leq$ -5 mA		-300 -450	+300 +450	mV mV
Standby current drain	I_{SCD}	V_{IN} = 20 V V_{IN} = 35 V	I_L = -5 mA I_L = -5 mA		-7.0 -8.0	-0.5 -0.5	mA mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	18.5 V $\leq V_{IN} \leq$ 35 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 20 V	-1.0 A $\leq I_L \leq$ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 20 V V_{IN} = 35 V			-3.50 -2.00	-0.02 -0.02	A A
Overload current 6/	I_{OL}	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		See figure 9	-4.0	-1.0	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 08 4/ (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 20 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	See figure 10 T_A = +25°C	53		dB
Output noise voltage 3/	N_o	V_{IN} = 20 V	I_L = -0.1 A	See figure 11 T_A = +25°C		300	μV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 20 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 20 V	I_L = -100 mA ΔI_L = -400 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 20 V	I_L = -5 mA	T_A = +125°C and -55°C	-3.75	+3.75	mV/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 09 4/ (see 3.4), unless otherwise specified.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Output voltage	V_{OUT}	V_{IN} = 28 V	I_L = -5 mA, -1.0 A		22.80	25.20	V
		V_{IN} = 38 V	I_L = -5 mA, -1.0 A		22.80	25.20	V
		V_{IN} = 40 V	I_L = -5 mA, -0.1 A		22.80	25.20	V
		V_{IN} = 30 V	I_L = -5 mA	T_A = +150°C	22.56	25.44	V
Line regulation	V_{RLINE}	28 V ≤ V_{IN} ≤ 40 V	I_L = -0.1 A		-720	+720	mV
		28 V ≤ V_{IN} ≤ 38 V	I_L = -0.5 A		-240	+240	mV
Load regulation	V_{RLOAD}	V_{IN} = 30 V	-1.0 A ≤ I_L ≤ -5 mA		-480	+480	mV
		V_{IN} = 40 V	-0.1 A ≤ I_L ≤ -5 mA		-720	+720	mV
Standby current drain	I_{SCD}	V_{IN} = 30 V	I_L = -5 mA		-7.0	-0.5	mA
		V_{IN} = 40 V	I_L = -5 mA		-8.0	-0.5	mA
Standby current drain change (versus line voltage)	ΔI_{SCD} (line)	28 V ≤ V_{IN} ≤ 40 V	I_L = -5 mA		-1.0	+1.0	mA
Standby current drain change (versus load current)	ΔI_{SCD} (load)	V_{IN} = 30 V	-1.0 A ≤ I_L ≤ -5 mA		-0.5	+0.5	mA
Output short circuit current 2/ 5/	I_{OS}	V_{IN} = 30 V			-2.50	-0.02	A
		V_{IN} = 35 V			-2.00	-0.005	A
Overload current 6/	I_{OL}	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		See figure 9	-4.0	-1.0	A

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 09 4/ (see 3.4), unless otherwise specified) - Continued.

Characteristics	Symbol	Conditions (see figure 8, unless otherwise specified)			Limits		Unit
		Input voltage	Load current	Other	Min	Max	
Ripple rejection 3/	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	V_{IN} = 30 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	See figure 10 T_A = +25°C	50		dB
Output noise voltage 3/	N_O	V_{IN} = 30 V	I_L = -0.1 A	See figure 11 T_A = +25°C		500	µV rms
Line transient response	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	V_{IN} = 30 V V_{pulse} = 3.0 V	I_L = -5 mA	See figure 12 T_A = +25°C		30	mV/V
Load transient response	$\frac{\Delta V_{OUT}}{\Delta I_L}$	V_{IN} = 30 V	I_L = -100 mA ΔI_L = -400 mA	See figure 13 T_A = +25°C		2.5	mV/mA
Average temperature coefficient of output voltage	$\frac{\Delta V_{OUT}}{\Delta T}$	V_{IN} = 30 V	I_L = -5 mA	T_A = +125°C and -55°C	-6.0	+6.0	mV/°C

1/ V_{OUT} test at V_{IN} = 8 V is conducted at T_A = -55°C only.

2/ One second maximum test duration.

3/ The meter for e_i and e_o shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.

4/ For device types 02 through 09, all tests performed at T_A = +125°C may, at the manufacturer's option, be performed at T_A = +150°C. Specifications for T_A = +125°C shall then apply at T_A = +150°C.

5/ A 1 µF capacitor may be added to dampen oscillations during the I_{OS} test for device types 02 through 09.

6/ The overload current test checks the device current limiting, and it is not a measure of the output peak current.

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3.4.2 Test limits. The test limits specified in tables I and III apply only for the stated test conditions (e.g., two percent duty cycle), which essentially keep the junction temperature constant. In most applications the junction temperature will greatly exceed the +25°C ambient or sink temperature; thus devices may not perform within the +25°C specified limits.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,4	1*,2,3,4
Group A test requirements (method 5005)	1,2,3,4,7,8	1,2,3,4,7,8
Group C end-point and group B, class S, electrical parameters (method 5005)	1,2,3 and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1,2,3	1

* PDA applies to subgroup 1 (see 4.3e).

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by PIN. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 52 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1, 4.4.2, 4.4.3, and 4.4.4).

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. For class S and class B devices, an additional burn-in screen shall be performed in accordance with figure 6 to test the operation of the thermal shutdown circuit. This screen shall be performed after serialization (method 5004 of MIL-STD-883) and before interim electrical parameters (method 5004 of MIL-STD-883). The requirements of 3.2.3 of method 1015 of MIL-STD-883 shall apply to this screen except the devices need not be tested in an oven.

b. Interim electrical parameters shall be performed for class S and class B devices and shall be as specified in table III.

c. Burn-in (method 1015 of MIL-STD-883). For class S and class B devices, only test condition C shall apply using the circuit shown on figure 7. Accelerated test conditions shall not be used.

d. Reverse bias burn-in shall not be performed.

e. Percent defective allowable (PDA). The PDA for class S and class B devices shall be as specified in MIL-M-38510, based on the failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

f. Constant acceleration (method 2001 of MIL-STD-883). Test condition B shall be used for case Y.

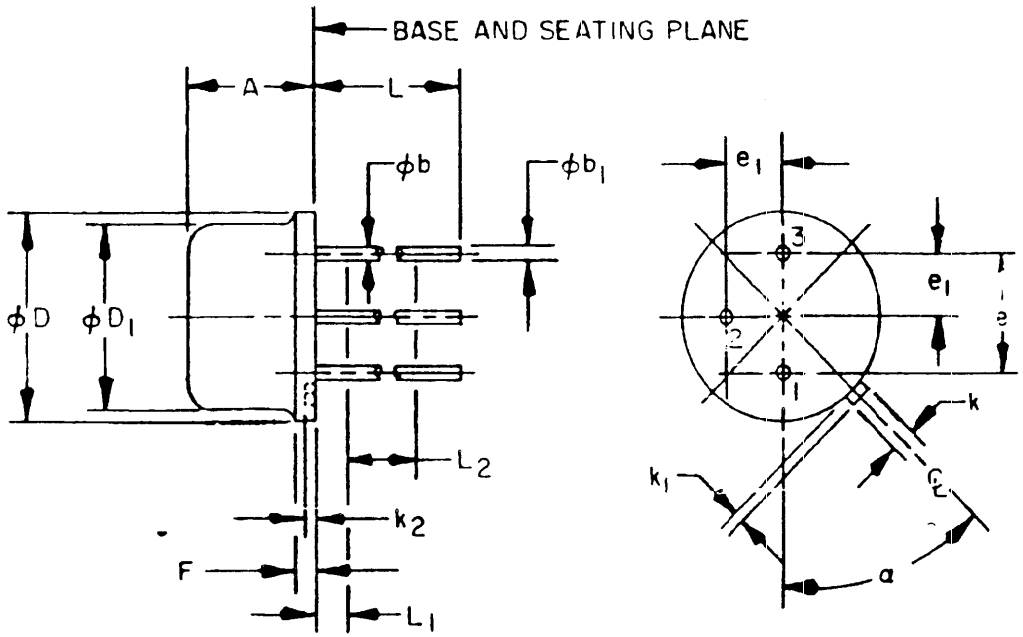
4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1, 4.4.2, 4.4.3, and 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 5, 6, 9, 10, and 11 shall be omitted.

b. Tests shall be as specified in table II herein.

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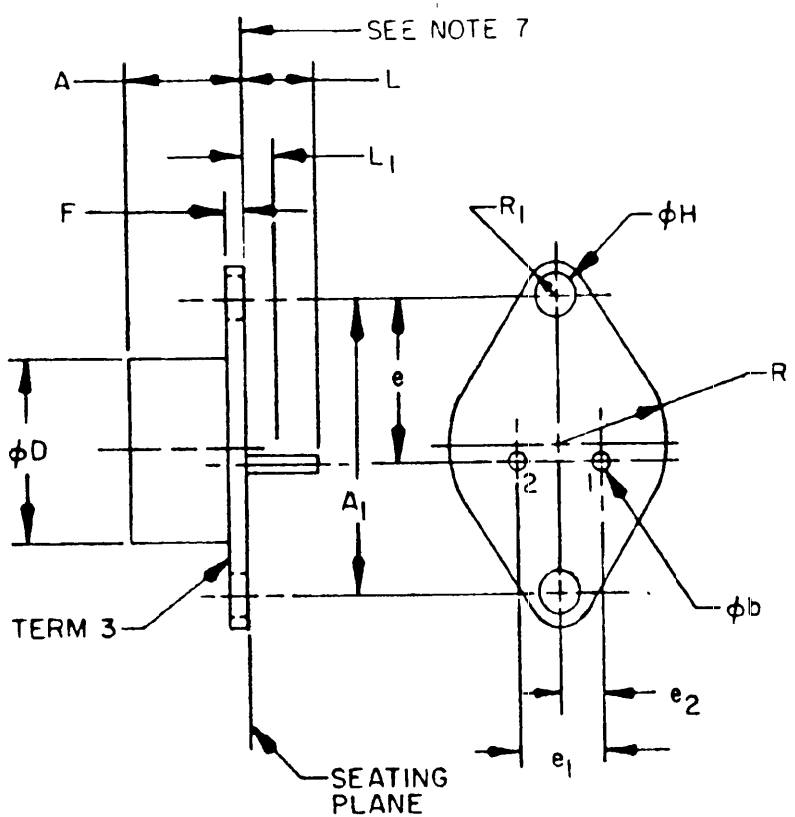


Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.165	.185	4.19	4.70	
øb	.016	.019	0.41	0.48	3
$øb_1$	.016	.021	0.41	0.53	3
øD	.335	.370	8.51	9.40	
$øD_1$	.305	.335	7.75	8.51	
e	.200 T.P.		5.08 T.P.		5
e_1	.100 T.P.		2.54 T.P.		5
F		.050		1.27	
k	.028	.034	0.71	0.86	
k_1	.029	.045	0.74	1.14	4
k_2	.009	.041	0.23	1.04	
L	.500		12.70		
L_1		.050		1.27	
L_2	.250		6.35		
α	45° T.P.		45° T.P.		5

NOTES:
1. Dimensions shall be measured in inches.
2. Metric equivalents are given for general information only.
3. (All leads)øb applies between L_1 and L_2. $øb_1$ applies between L_2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the reference plane.
4. Measured form the maximum diameter of the product.
5. Leads having a maximum diameter .019 (0.48 mm) measured in gauging plane .054 (1.37 mm) + .001 (0.03 mm) - .000 (0.00 mm) below the base plane of the product shall be within .007 (0.18 mm) of their true position relative to a maximum width tab.
6. The product may be measured by direct methods or by gauge.

FIGURE 1. Case outline X.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.450	6.35	11.43	
A_1	1.177	1.197	29.90	30.40	
øb	.038	.043	0.97	1.09	3,7
øD		.875		22.22	
e	.655	.675	16.64	17.14	
e_1	.420	.440	10.67	11.18	
e_2	.205	.225	5.21	5.72	
F	.060	.135	1.52	3.43	
øH	.151	.161	3.84	4.09	5,6
L	.312	.500	7.92	12.70	4
L_1		.050		1.27	3,4
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	

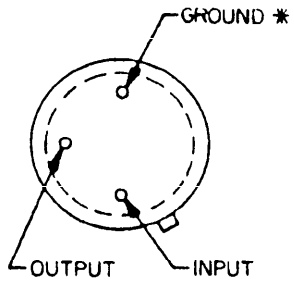
NOTES:
1. Dimensions shall be measured in inches.
2. Metric equivalents are given for general information only.
3. (Two leads) øb applies between L_1 and .500 (12.70 mm) from the seating plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the seating plane.
4. Two leads.
5. Two holes.
6. Two holes located at true position within diameter .010 (0.25 mm).
7. Leads having a maximum diameter of .043 (1.09 mm)measured in gauging plane .054 (1.37 mm) +.001 (0.03 mm) -.000 (0.00 mm) below the seating plane shall be located at true position within diameter .014 (0.36 mm).
8. The mounting surface of the header shall be flat to convex within .003 (0.08 mm) inside a .930 (23.62 mm) diameter circle on the center of the header and flat to convex within .006 (0.15 mm) overall.

FIGURE 2. Case outline Y.

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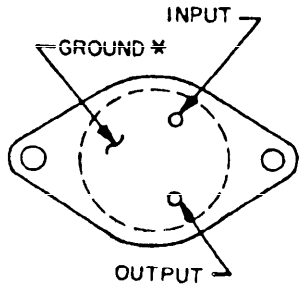
Device types 01, 02, 03, 04, and 05

Case X



Device types 01, 06, 07, 08, and 09

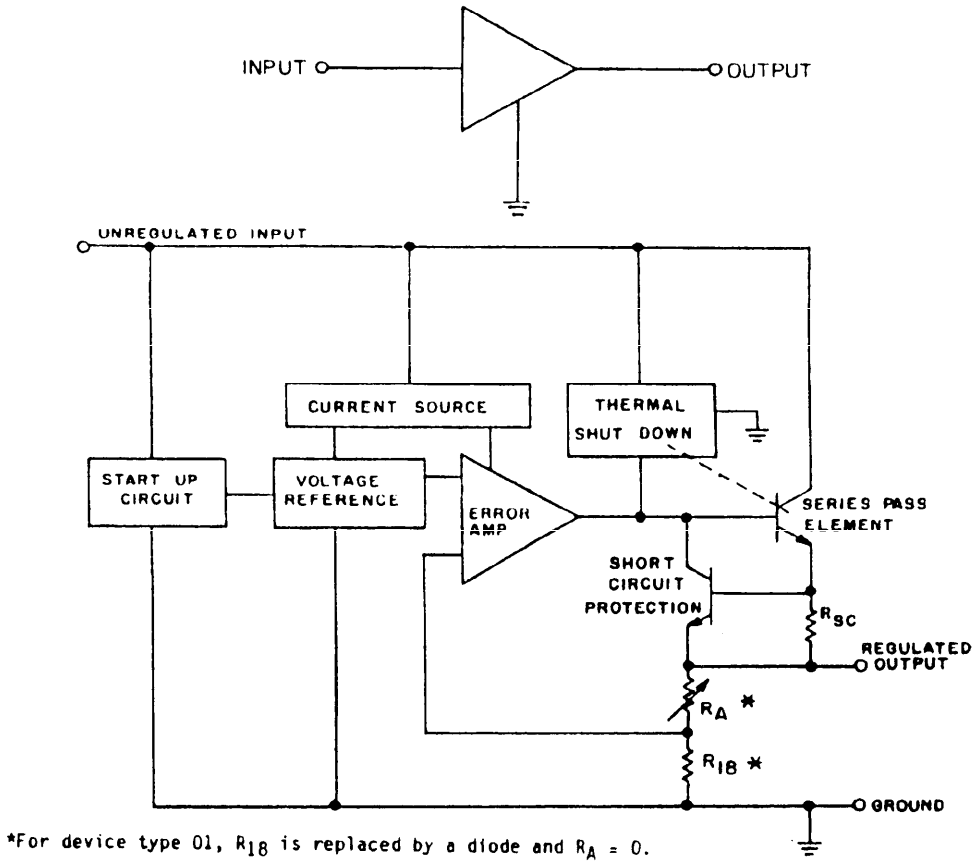
Case Y



*(case is connected to ground.)

NOTE: Bottom views.

FIGURE 3. Terminal connections.



*For device type 01, R_{1B} is replaced by a diode and R_A = 0.

FIGURE 4. Block diagram.

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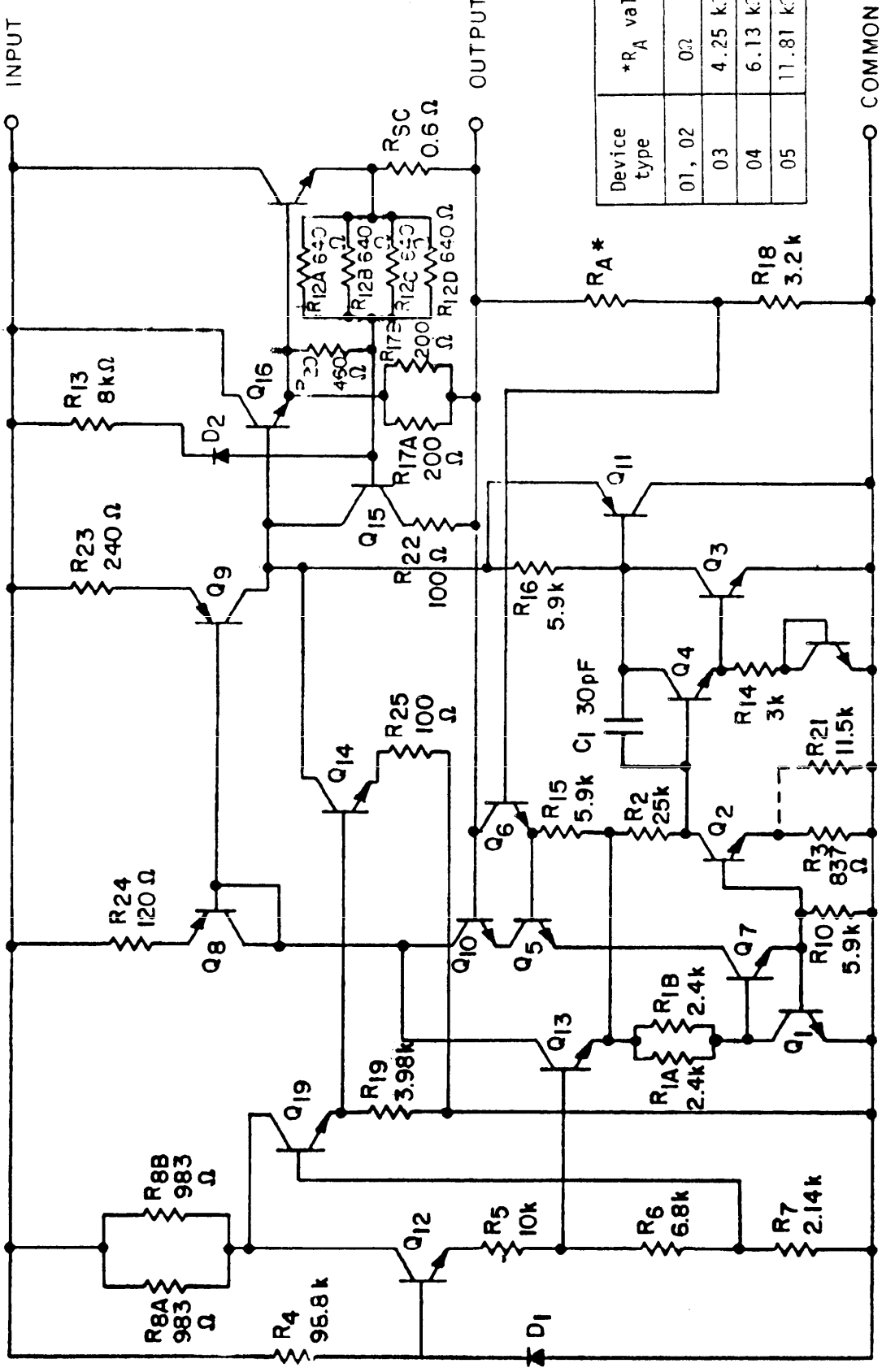
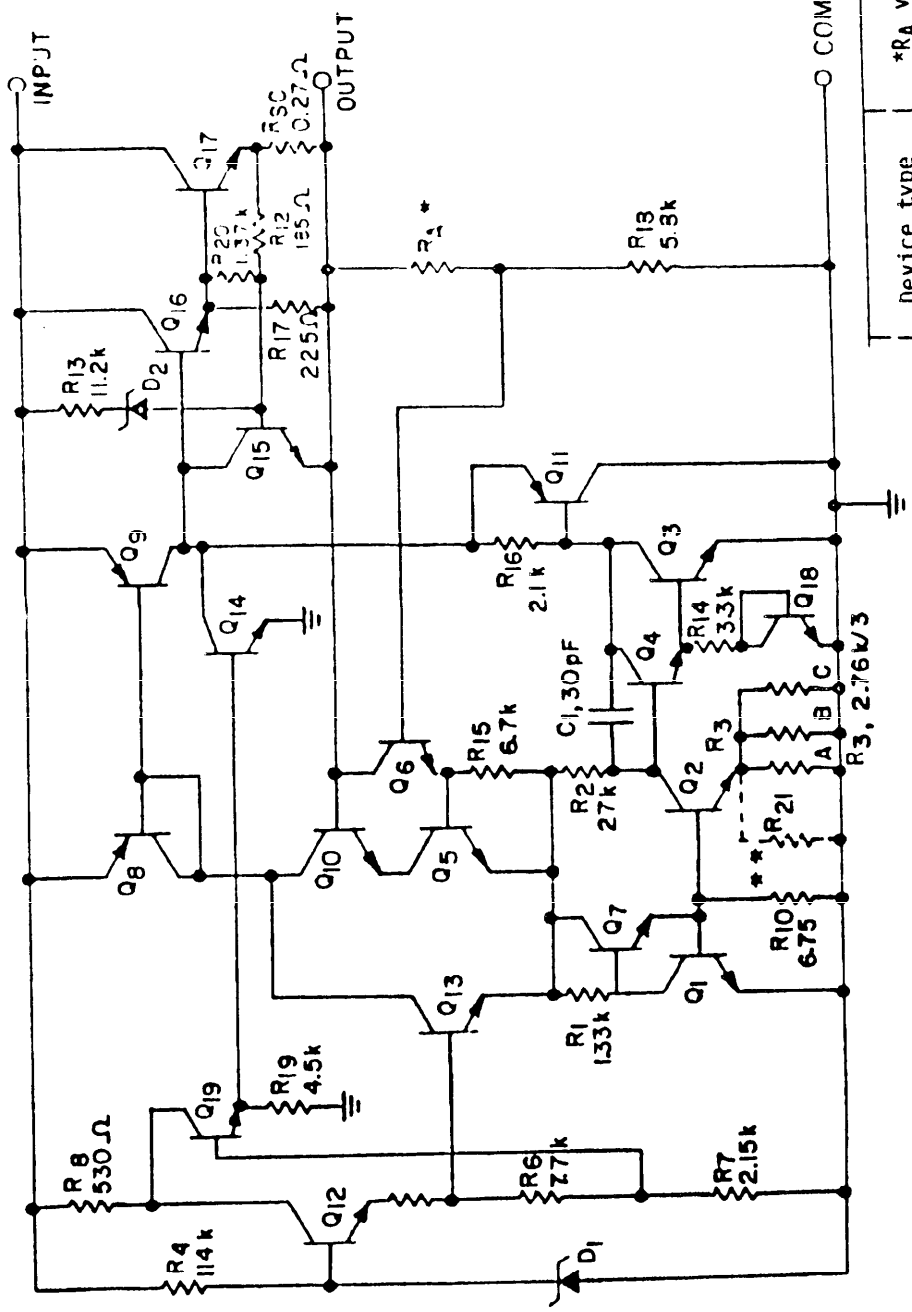


FIGURE 5. Functional schematic circuits.

** R21 not connected in device types 01 and 02.

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Device types 06, 07, 08, and 09

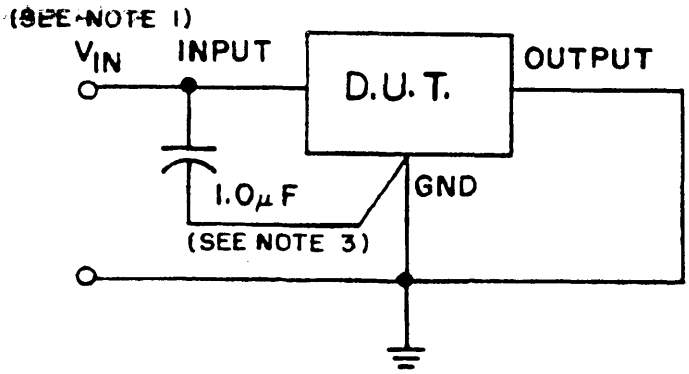


Device type	*R_A value
06	0Ω
07	7.575 kΩ
08	10.875 kΩ
09	20.875 kΩ

**Added for $V_{OUT} \geq 6$ V R21, 13.3 kΩ. Not connected in device type 06.

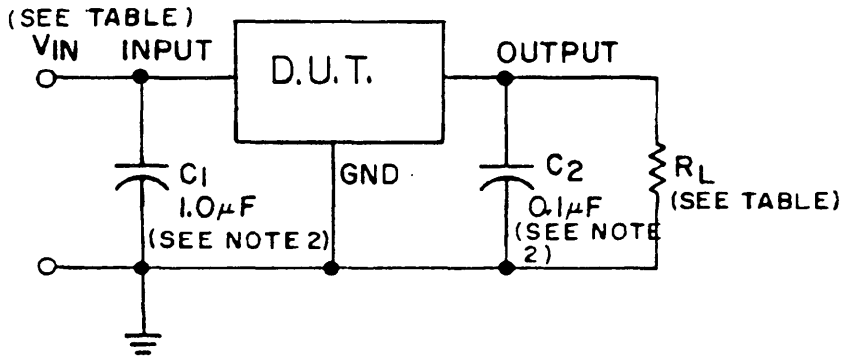
FIGURE 5. Functional schematic circuits - Continued.

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NOTES:
1. V_{IN} = 20 V, for device types 01, 02, and 06. V_{IN} = 25 V for device types 03, 04, 05, 07, 08, and 09.
2. Test is conducted at T_A = 25°C without a heat sink for a minimum of 30 minutes.
3. The 1.0 μF capacitor may be reduced or eliminated provided the device remains stable.

FIGURE 6. Pre burn-in test circuit.



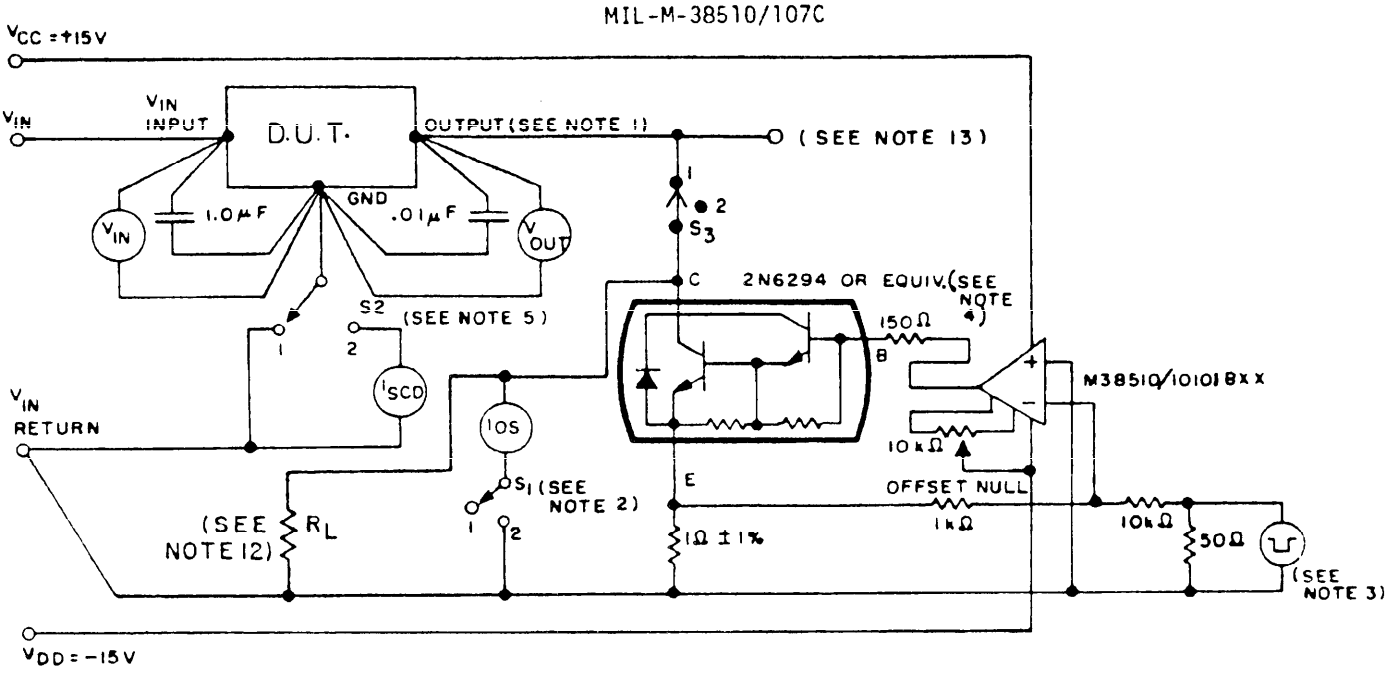
Device table.

Device type	01	02	03	04	05	06	07	08	09
V_{IN}	25 V	25 V	30 V	30 V	31 V	25 V	30 V	30 V	33 V
R_L	See note 3	1.6 kΩ	6.8 kΩ	6.8 kΩ	6.8 kΩ	137Ω	316Ω	316Ω	316Ω

NOTES:
1. Test is conducted without a heat sink.
2. The 1.0 μF capacitor and/or the 0.1 μF capacitor may be reduced or eliminated provided the device remains stable.
3. For case outline X, R_L = 2 kΩ ±5% and for case outline Y, R_L = 680Ω ±5%.

FIGURE 7. Burn-in test and operating life test circuit.

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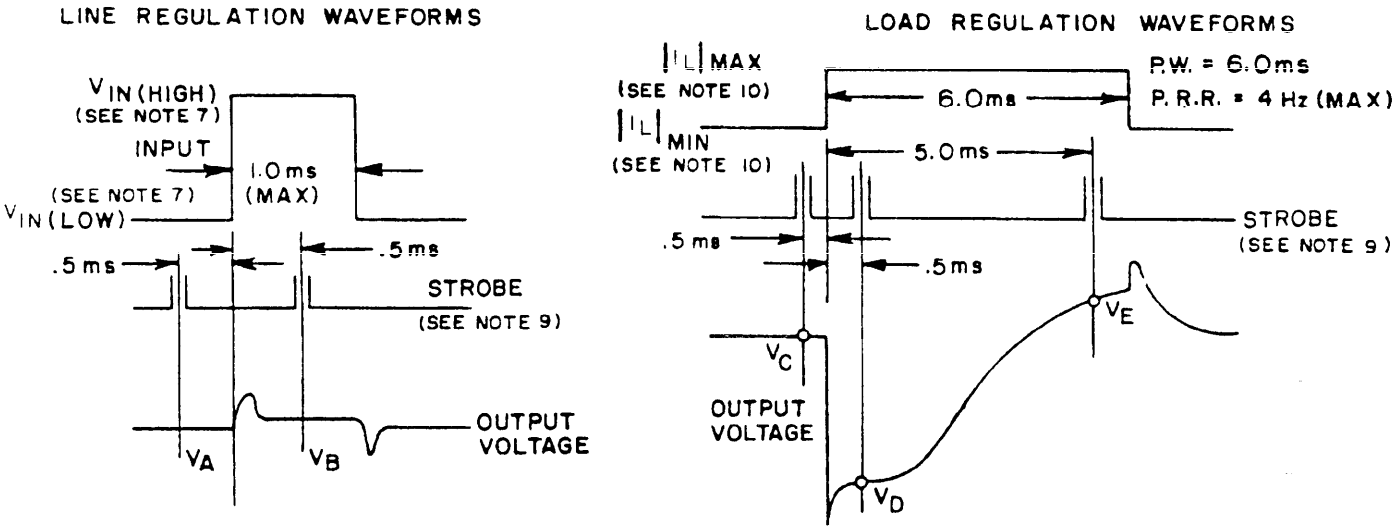


FIGURE 8. Test circuit for static tests for device types 02, 03, 04, 05, 06, 07, 08, and 09.

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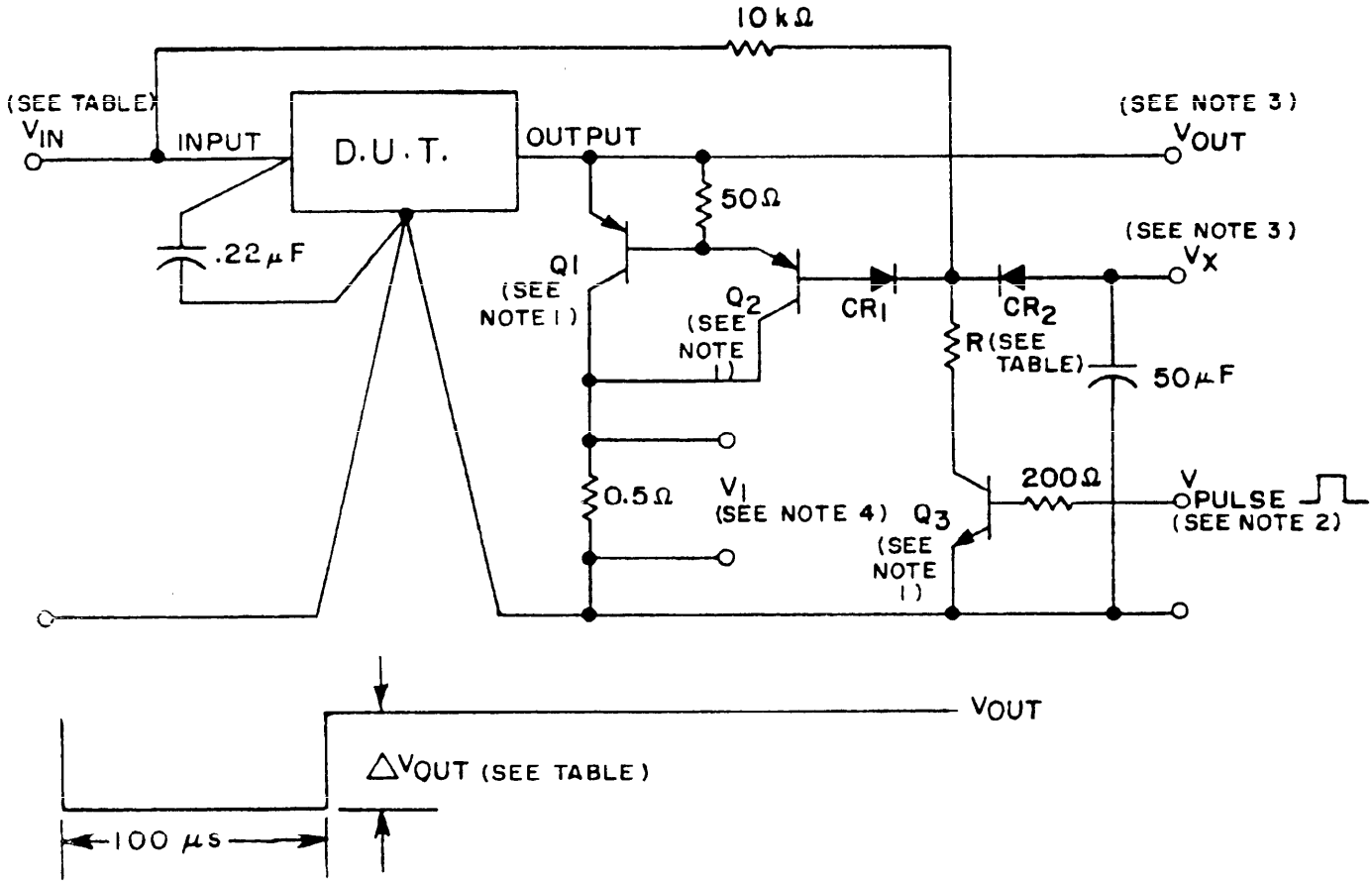
NOTES:
1. Output terminal must utilize a kelvin connection.
2. I_{OS} test duration: 1 second maximum.
   S_1: Position 2 for I_{OS} test.
   Position 1 for all other tests.
3. The pulse generator has the following characteristics:
   a. Pulse amplitude = -10 I_L volts.
   b. Pulse width = 1.0 ms.
   c. Duty cycle = 2 percent (maximum).
   Load current is determined by the voltage across the 1Ω resistor.
   Measurements shall be made 0.5 ms after the start of the pulse.
4. An alternate drive circuit for the 2N6294 transistor may be used to develop the proper load current through the 1Ω resistor.
5. Use position 2 of S_2 for I_{SCD}, ΔI_{SCD} tests only.
6. Thermal shutdown output voltage test is conducted without a heat sink at an ambient temperature of T_A = +150°C.
7. $V_{IN(LOW)}$ and $V_{IN(HIGH)}$ in accordance with table III.
8. $V_{RLINE} = V_B - V_A$.
9. The output voltage is sampled at the specified intervals. Strobe pulse width is 100 µs maximum.
10. $|I_L|$ minimum and $|I_L|$ maximum in accordance with table III.
11. $V_{RLOAD} = V_D - V_C$ and $V_{RLOAD} = V_E - V_C$.
12.

Device type	02	03	04	05	06	07	08	09
R_L	1 kΩ	2.4 kΩ	3 kΩ	4.8 kΩ	1 kΩ	2.4 kΩ	3 kΩ	4.8 kΩ

13. Alternate point for applying load circuit directly with automatic test equipment (place S3 in position 2).
14. A µF capacitor may added to dampen oscillations during the I_{OS} test for device types 02 through 09.

FIGURE 8. Test circuit for static tests for device types 02, 03, 04, 05, 06, 07, 08, and 09 - Continued.

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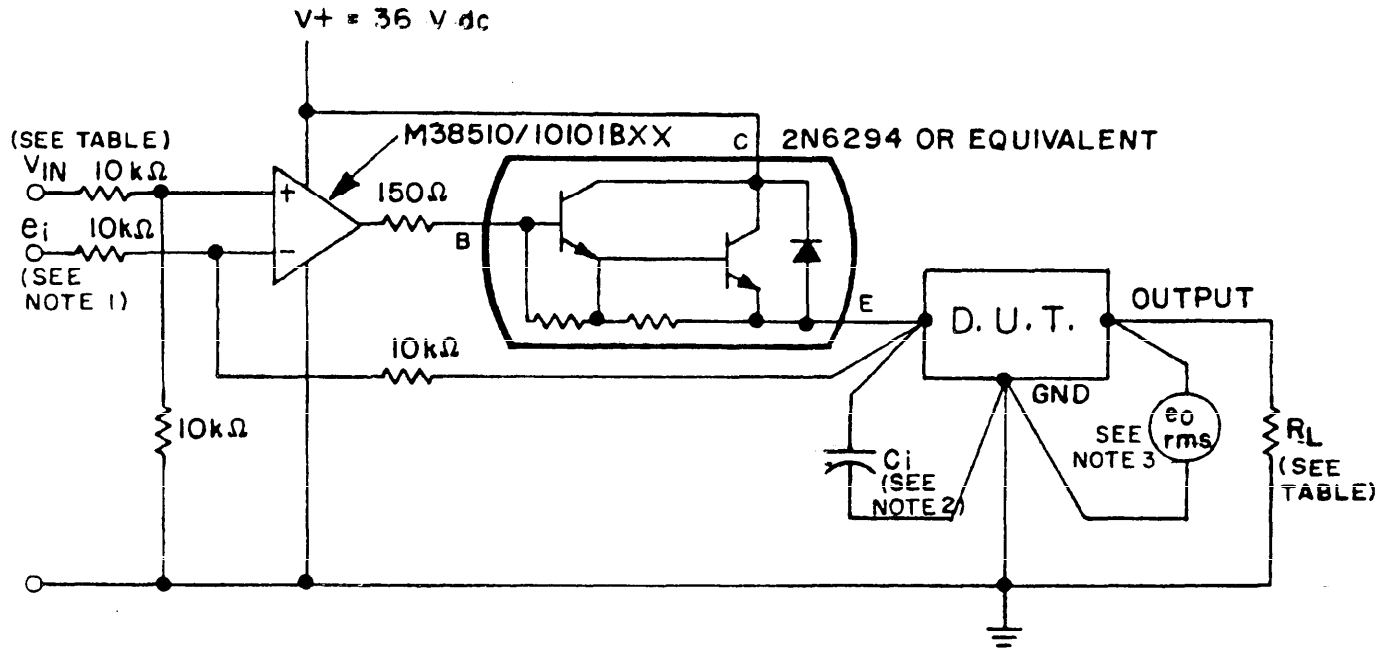
Device table.

Device type	02	03	04	05	06	07	08	09
V_{IN}	8 V	15 V	18.5 V	28 V	8 V	15 V	18.5 V	28 V
ΔV_{OUT}	-0.48 V	-1.13 V	-1.43 V	-2.28 V	-0.48 V	-1.13 V	-1.43 V	-2.28 V
R	110Ω	300Ω	390Ω	680Ω	110Ω	300Ω	390Ω	680Ω

NOTES:
1. Q_1 = 2N6187 or equivalent.
   Q_2 = 2N4033 or equivalent.
   Q_3 = 2N3108 or equivalent.
   CR_1 = 1N3873 or equivalent.
   CR_2 = 1N3873 or equivalent.
2. The pulse, V_{pulse}, has the following characteristics:
   Pulse width = 100 μs.
   Pulse repetition rate = 200 Hz.
   Pulse amplitude = 5 V.
3. Adjust V_X for a forced output voltage condition during application of V_{pulse}.
4. Kelvin connections shall be made directly to the 0.5Ω resistor. The overload current is I_{OL} = 2 (V_I) amperes.
5. The overload current test checks the device current limiting, and is not a measure of the output peak current.

FIGURE 9. Overload current test circuit for device types 02, 03, 04, 05 06, 07, 08, and 09.

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Device table.

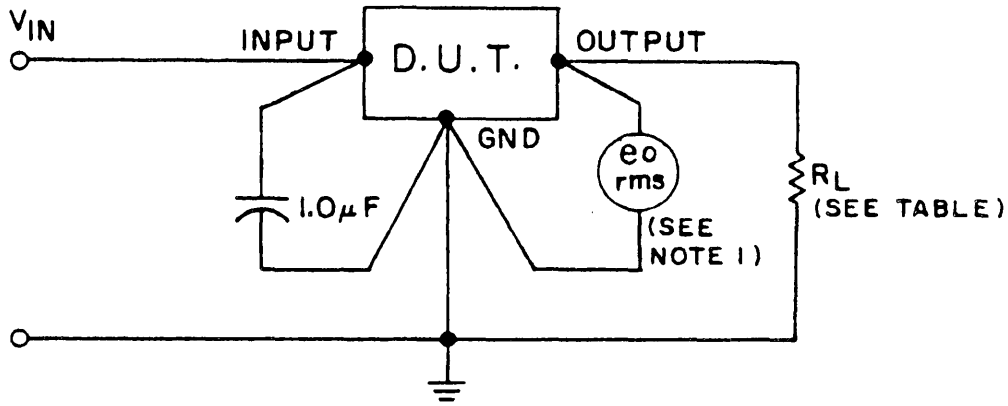
Device type	01	02	03	04	05	06	07	08	09
V_{IN}	10 V	10 V	17 V	20 V	30 V	10 V	17 V	20 V	30 V
R_L	40.2Ω*	40.2Ω	95.3Ω	121Ω	191Ω	14.3Ω	34Ω	43.2Ω	68.1Ω

R_L shall be type NER 70 or equivalent.
*This value shall be used for case X. For case Y use R_L = 14.3Ω.

NOTES:
1. e_i = 1 V rms at f = 2400 Hz (measured at the input terminals of D.U.T.).
   Ripple rejection (db) = $20 \log \left(\frac{e_i \text{ rms.}}{e_o \text{ rms}} \right)$
2. c_i shall be minimum value to prevent oscillations of the device under test. The maximum value of c_i shall be 0.4 µF.
3. The meter for e_i and e_o shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.

FIGURE 10. Ripple rejection test circuit for all device types.

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Device table.

Device type	01	02	03	04	05	06	07	08	09
V_{IN}	10 V	10 V	17 V	20 V	30 V	10 V	17 V	20 V	30 V
R_L	100Ω*	100Ω	237Ω	301Ω	475Ω	49.9Ω	121Ω	150Ω	237Ω

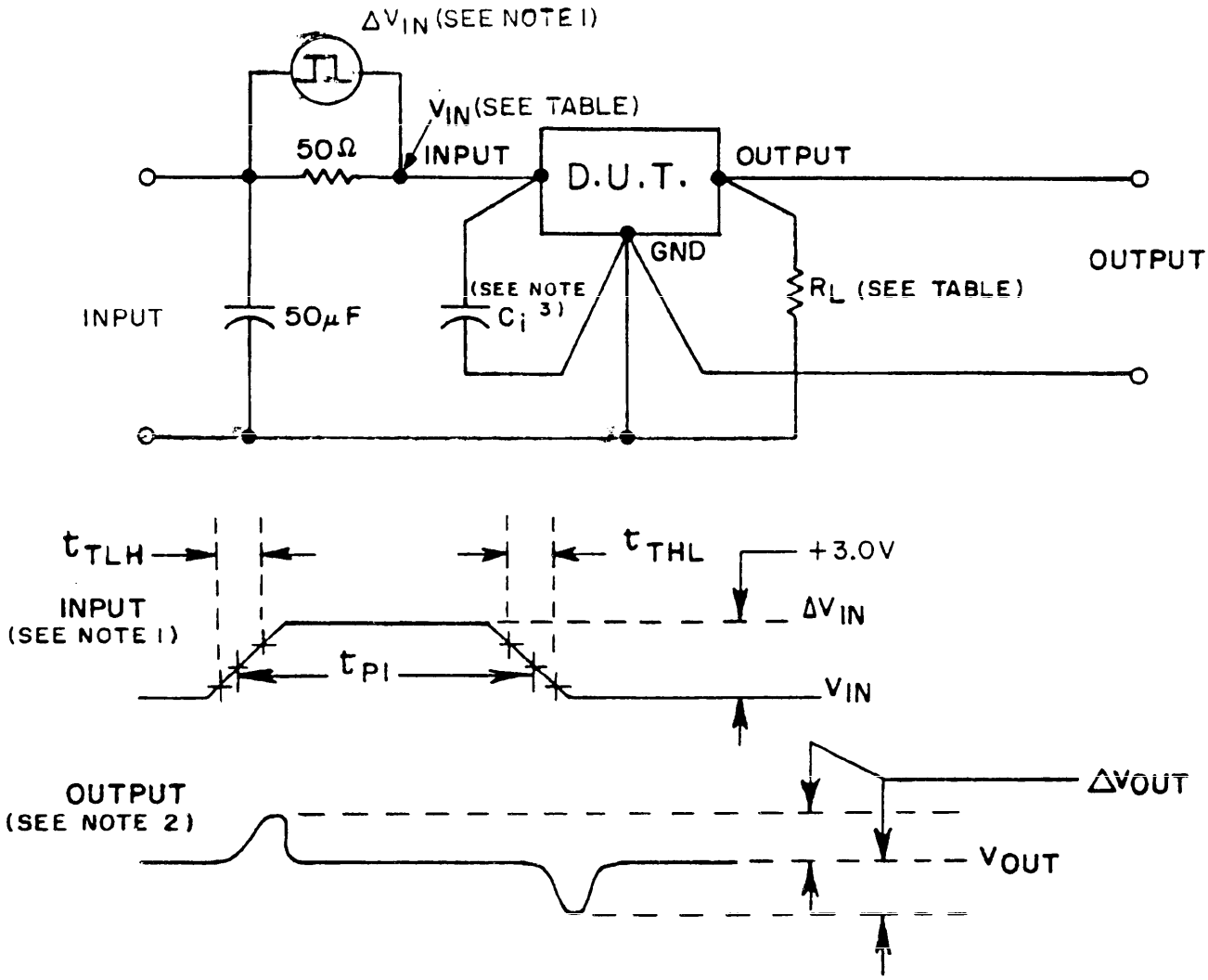
R_L shall be type RER 70 or equivalent.

*This value shall be used for case X. For case Y use R_L = 49.9Ω.

NOTES:
1. The meter for measuring e_o rms shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.
2. N_o = e_o rms.

FIGURE 11. Noise test circuit for all device types.

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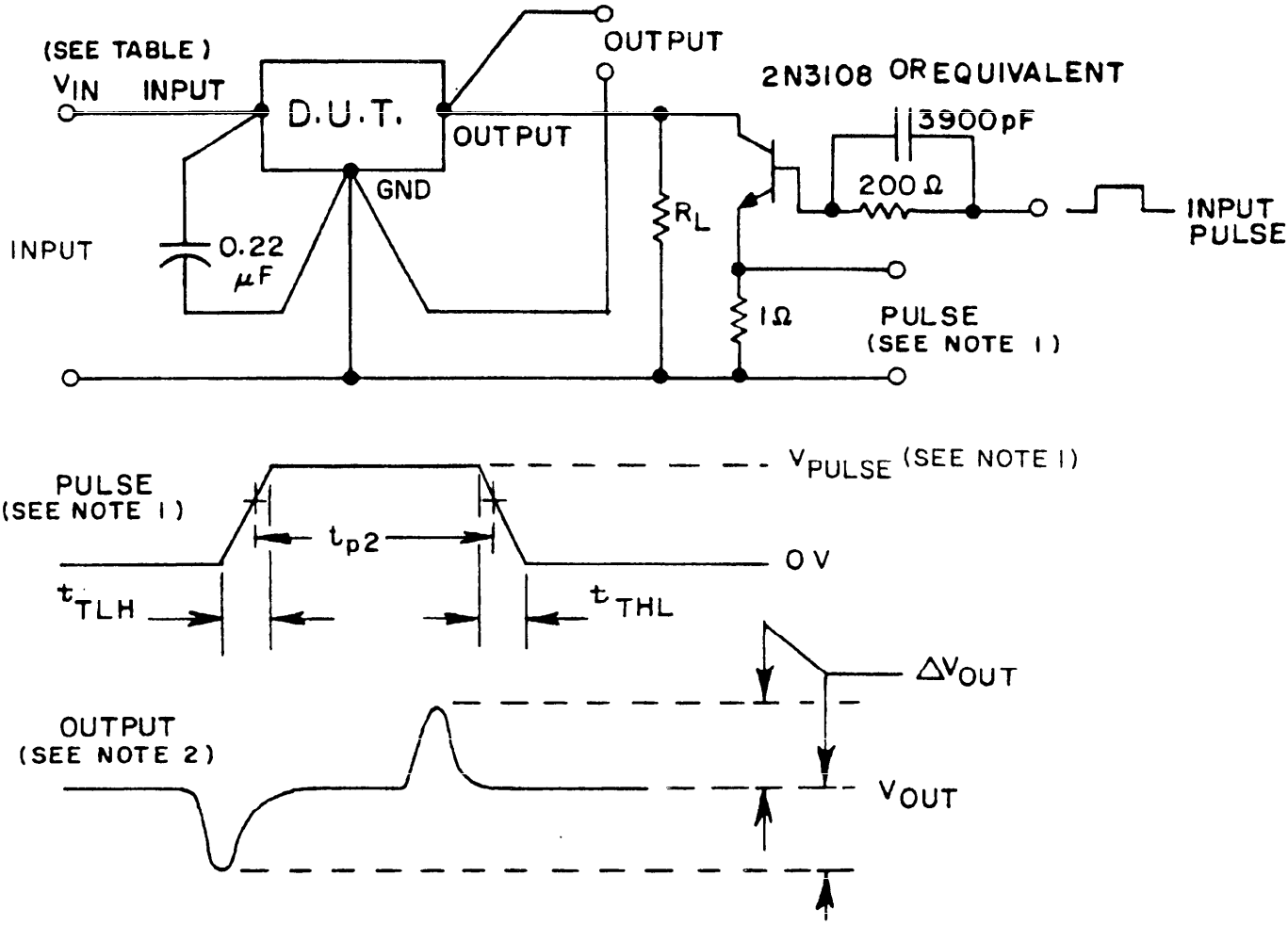
Device type	01,02,06	03,07	04,08	05,09
V_{IN}	10 V	17 V	20 V	30 V
*R_L	1.0 kΩ	2.4 kΩ	3.0 kΩ	4.8 kΩ

*R_L shall be type RER 70 or equivalent.

NOTES:
1. The pulse characteristics shall be established under load while the D.U.T. is operating. $t_{THL} = t_{TLH} = 5.0$ μs.
   $\Delta V_{IN} = V_{IN} + 3.0$ V
   $t_{PI} = 25$ μs at 3 percent duty cycle.
2. $\Delta V_{OUT} = 90$ mV maximum (this guarantees the specification limit of 30 mV/V).
3. C_i shall be minimum value to prevent device oscillations.

FIGURE 12. Line transient response test circuit for all device types.

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Device type	01	02	03	04	05	06	07	08	09
V_{IN}	10 V	10 V	17 V	20 V	30 V	10 V	17 V	20 V	30 V
*R_L	100Ω**	100Ω	243Ω	301Ω	487Ω	49.9Ω	121Ω	150Ω	243Ω

*Load resistor R_L is type RER 70 or equivalent.

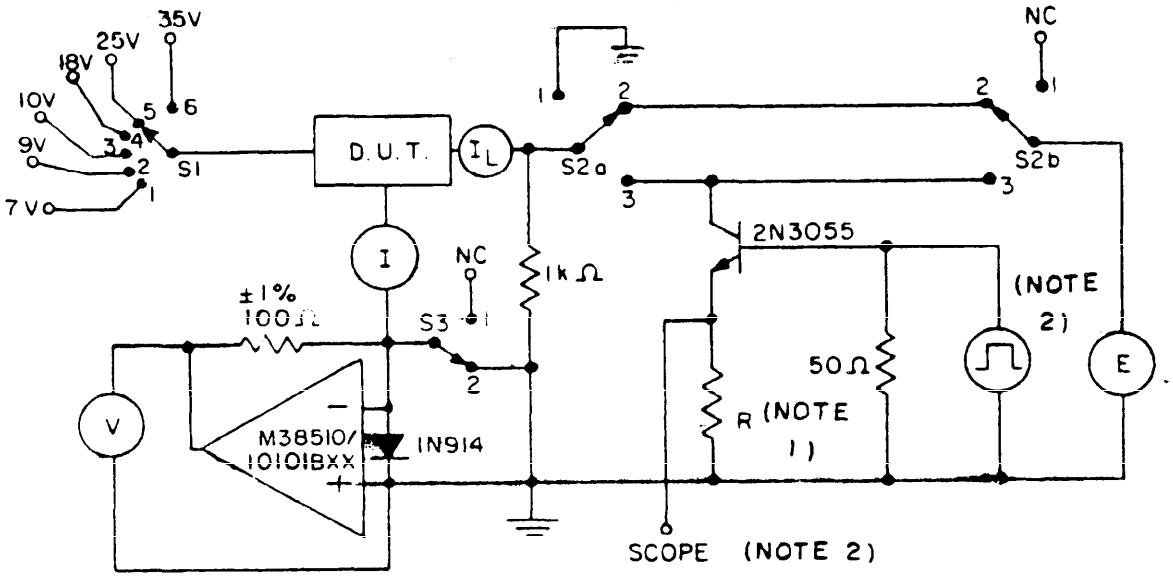
**This value shall be used for case X. For case Y use R_L = 49.9Ω.

NOTES:

1. The pulse characteristics shall be established under load while the D.U.T. is operating.
   $t_{THL} = t_{TLH}$ = 1.0 μs.
   t_{p2} = 25 μs at 3 percent duty cycle.
   V_{PULSE} = 0.20 V for device types 01 (case X), 02, 03, 04, and 05.
   V_{PULSE} = 0.40 V for device types 01 (case Y), 06, 07, 08, and 09.
2. ΔV_{OUT} = 400 mV maximum for device type 01 (case X).
   ΔV_{OUT} = 800 mV maximum for device type 01 (case Y).
   ΔV_{OUT} = 500 mV maximum for device types 02 - 05.
   ΔV_{OUT} = 1000 mV maximum for device types 06 - 09.
   (These values guarantee the specified limits for load transient response.)
3. Oscilloscope bandwidth shall be greater than 5 MHz.

FIGURE 13. Load transient response test circuit for all device types.

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Parameter	Test	I_L (mA)	Switch positions			Measure		Measured parameters equation	Units
			S1	S2	S3	Value	Units		
V_{OUT}	1, 14, 27	-5	1	3	2	E_1	V	$V_{OUT} = E_1$	V
	2, 15, 28	See notes 3 and 7	1	3	2	E_2	V	$V_{OUT} = E_2$	V
	3, 16, 29	See note 3	2	3	2	E_3	V	$V_{OUT} = E_3$	V
	4, 17, 30	See note 7	4	3	2	E_4	V	$V_{OUT} = E_4$	V
	5, 18, 31	-5	5	3	2	E_5	V	$V_{OUT} = E_5$	V
	6, 19, 32	See note 8	5	3	2	E_6	V	$V_{OUT} = E_6$	V
V_{RLINE}	7, 20, 33	-5	1	2	2	E_7	V	$V_{RLINE} = \frac{E_8-E_7}{1,000}$	mV
		-5	5	2	2	E_8	V		
V_{LOAD}	8, 21, 34	-5	3	3	2	E_9	V	$V_{LOAD} = \frac{E_{10}-E_9}{1,000}$	mV
		See notes 3 and 7	3	3	2	E_{10}	V		

FIGURE 14. Test circuit for static and dynamic tests for device type 01.

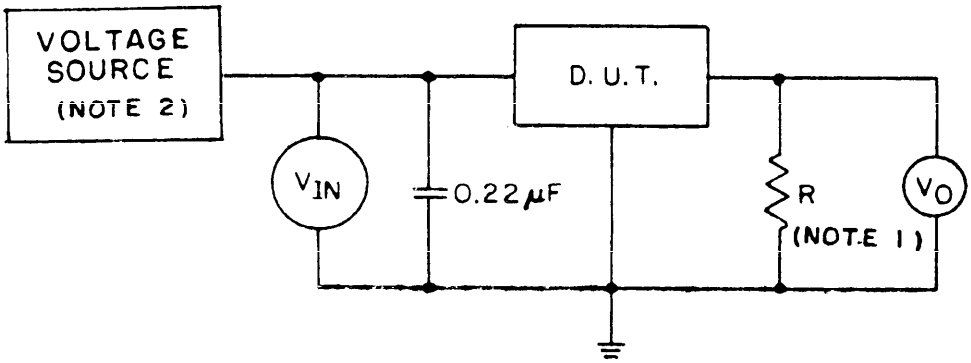
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Parameter	Test	I_L (mA)	Switch positions			Measure		Measured parameters equation	Units
			S1	S2	S3	Value	Units		
I_{SCD}	9, 22, 35	-5	1	2	2	I_1	mA	$I_{SCD} = I_1$	mA
		-5	5	2	2	I_2	mA	$I_{SCD} = I_2$	
ΔI_{SCD} (line)	10, 23, 36	-5	1	2	1	V_1	V	$\Delta I_{SCD} = IO(V_2-V_1)$ (line)	mA
		-5	5	2	1	V_2	V		
ΔI_{SCD} (load)	11, 24, 37	-5	3	3	1	V_3	V	$\Delta I_{SCD} = IO(V_4-V_3)$ (load)	mA
		See notes 3 and 7	3	3	1	V_4	V		
I_{OS}	12, 25, 38	See note 6	6	1	2	I_L	A	$I_{OS} = I_L$	A

NOTES:
1. R = 2Ω ±1 percent for case X; R = 1Ω for case Y.
2. The pulse generator has the following characteristics: Variable amplitude, pulse width = 50 µs and PRR = 200 Hz. Pulse amplitude is adjusted to set the load current by monitoring the voltage drop across R.
3. I_L = -0.5 A for case X.
4. $+25^{\circ}C \leq T_A \leq +125^{\circ}C$; $\Delta V_{OUT/\Delta T}$ = (V_{OUT} at +25°C) - (V_{OUT} at +125°C) x 10.
5. $+55^{\circ}C \leq T_A \leq +25^{\circ}C$; $\Delta V_{OUT/\Delta T}$ = (V_{OUT} at -55°C) - (V_{OUT} at +25°C) x 12.5.
6. One second maximum test duration.
7. I_L = 1.5 A for case Y.
8. I_L = -100 mA for case X; I_L = -1.0 A for case Y.

FIGURE 14. Test circuit for static and dynamic tests for device type 01 - Continued.

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NOTES:
1. R = 25Ω ±5 percent for case X and 5Ω ±5 percent for case Y.
2. Voltage source shall be a variable power supply capable of producing the following input waveform. Device shall turn on at $V_{IN} \leq 9$ V and shall remain on when the input is returned to 7 V.

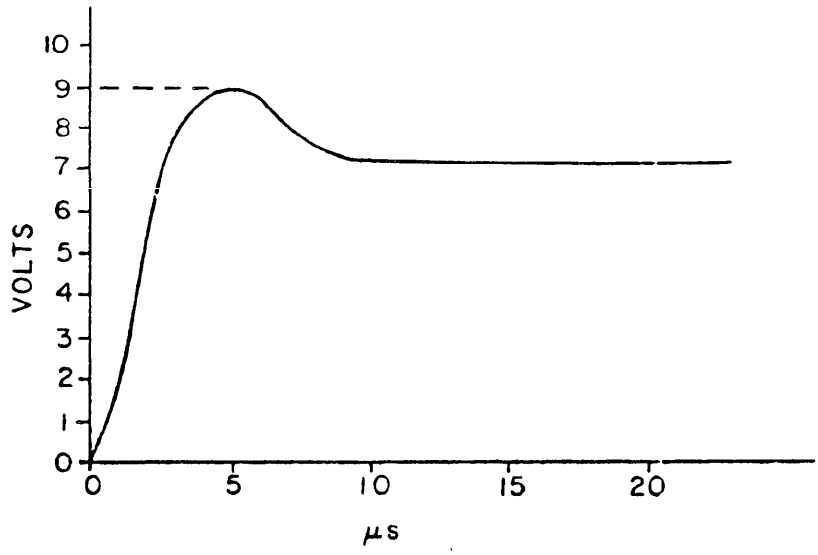


FIGURE 15. Start-up input voltage test circuit for device type 01.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	Conditions (figure 14, unless otherwise specified) Input voltage	Load current I_L for case X	Load current I_L for case Y	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 7 V	-5 mA	-5 mA	4.70	5.40	V	
		2	V_{IN} = 7 V	-0.5 A	-1.5 A	"	"	"	
		3	V_{IN} = 9 V	-0.5 A		"	"	"	
		4	V_{IN} = 18 V		-1.5 A	"	"	"	
		5	V_{IN} = 25 V	-5 mA	-5 mA	"	"	"	
		6	V_{IN} = 25 V	-100 mA	-1.0 A	"	"	"	
	V_{RLINE}	7	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-50	+50	mV	
	V_{RLOAD}	8	V_{IN} = 10 V	I_{MAX} ≤ I_L ≤ -5 mA		-100	+100	mV	
				I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
	I_{SCD}	9	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-10.0	+0.5	mA	
	ΔI_{SCD} (line)	10	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-0.5	+0.5	"	
	ΔI_{SCD} (load)	11	V_{IN} = 10 V	I_{MAX} ≤ I_L ≤ -5 mA		-0.8	+0.8	"	
				I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
	I_{OS}	12	V_{IN} = 35 V			-2.0	0.01	A	Case X 1/
						-2.8	0.01	A	Case Y 1/
	V_{START}	13		Case X (R_L = 25Ω ±5%)			9.0	V	See figure 15
				Case Y (R_L = 5Ω ±5%)					

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 14, unless otherwise specified) Input voltage	Load current I_L for case X	Load current I_L for case Y	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	14	V_{IN} = 7 V	-5 mA	-5 mA	4.70	5.40	V	
		15	V_{IN} = 7 V	-0.5 A	-1.5 A	"	"	"	
		16	V_{IN} = 9 V	-0.5 A		"	"	"	
		17	V_{IN} = 18 V		-1.5 A	"	"	"	
		18	V_{IN} = 25 V	-5 mA	-5 mA	"	"	"	
		19	V_{IN} = 25 V	-100 mA	-1.0 A	"	"	"	
	V_{RLINE}	20	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-100	+100	mV	
	V_{RLOAD}	21	V_{IN} = 10 V	$I_{MAX} \leq I_L \leq$ -5 mA		-100	+100	mV	
				I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
	I_{SCD}	22	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-10.0	+0.5	mA	
	ΔI_{SCD} (line)	23	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-0.5	+0.5	"	
	ΔI_{SCD} (load)	24	V_{IN} = 10 V	$I_{MAX} \leq I_L \leq$ -5 mA		-0.8	+0.8	"	
				I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
	I_{OS}	25	V_{IN} = 35 V			-2.0	0.01	A	Case X 1/
						-2.8	0.01	A	Case Y 1/
	V_{START}	26		Case X (R_L = 25Ω ±5%)			9.0	V	See figure 15
				Case Y (R_L = 5Ω ±5%)					

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 14, unless otherwise specified) Input voltage	Load current I_L for case X	Load current I_L for case Y	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	27	V_{IN} = 7 V	-5 mA	-5 mA	4.70	5.40	V	
		28	V_{IN} = 7 V	-0.5 A	-1.5 A	"	"	"	
		29	V_{IN} = 9 V	-0.5 A		"	"	"	
		30	V_{IN} = 18 V		-1.5 A	"	"	"	
		31	V_{IN} = 25 V	-5 mA	-5 mA	"	"	"	
		32	V_{IN} = 25 V	-100 mA	-1.0 A	"	"	"	
	V_{RLINE}	33	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-100	+100	mV	
	V_{RLOAD}	34	V_{IN} = 10 V	I_{MAX} ≤ I_L ≤ -5 mA		-100	+100	mV	
				I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
	I_{SCD}	35	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-10.0	+0.5	mA	
	ΔI_{SCD} (line)	36	7 V ≤ V_{IN} ≤ 25 V	-5 mA		-0.5	+0.5	"	
	ΔI_{SCD} (load)	37	V_{IN} = 10 V	I_{MAX} ≤ I_L ≤ -5 mA		-0.8	+0.8	"	
				I_{MAX} = -0.5 A	I_{MAX} = -1.5 A				
	I_{OS}	38	V_{IN} = 35 V			-2.0	0.01	A	Case X 1/
						-2.8	0.01	A	Case Y 1/
	V_{START}	39		Case X (R_L = 25Ω ±5%)			9.0	V	See figure 15
				Case Y (R_L = 5Ω ±5%)					

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Conditions (figure 14, unless otherwise specified) Input voltage	Load current I_L for case X	Load current I_L for case Y	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	40	V_{IN} = 10 V e_i = 1 V rms at f = 2,400 Hz	-125 mA	-350 mA	60		dB	See figure 10 2/
7 T_A = +25°C	N_o	41	V_{IN} = 10 V	-50 mA	-100 mA		125	μV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	42	V_{IN} = 10 V V_{pulse} = 3.0 V	-5 mA			15	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	43	V_{IN} = 10 V	-50 mA ΔI_L = -200 mA	-100 mA ΔI_L = -400 mA		2.0	mV/mA	See figure 13
8 T_A = +125°C	$\frac{\Delta V_{OUT}}{\Delta T}$	44	V_{IN} = 7 V	-5 mA		-2.0	+2.0	mV/°C	$\frac{\Delta V_{OUT}}{\Delta T} = (V_1 - V_{14}) \times 10$
8 T_A = -55°C	$\frac{\Delta V_{OUT}}{\Delta T}$	45	V_{IN} = 7 V	-5 mA		-2.0	+2.0	mV/°C	$\frac{\Delta V_{OUT}}{\Delta T} = (V_1 - V_{27}) \times 12.5$

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 8 V	I_L = -5 mA	4.75	5.25	V	
		2		I_L = -500 mA	"	"	"	
		3	V_{IN} = 20 V	I_L = -5 mA	"	"	"	
		4		I_L = -500 mA	"	"	"	
		5	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		6		I_L = -50 mA	"	"	"	
	V_{RLINE}	7	8 V ≤ V_{IN} ≤ 35 V	I_L = -50 mA	-150	+150	mV	See figure 8 waveforms
		8	8 V ≤ V_{IN} ≤ 25 V	I_L = -350 mA	-50	+50	"	
	V_{RLOAD}	9	V_{IN} = 10 V	-500 mA ≤ I_L ≤ -5 mA	-100	+100	"	
		10	V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA	-150	+150	"	
	I_{SCD}	11	V_{IN} = 10 V	I_L = -5 mA	-7.0	-0.5	mA	
		12	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	13	8 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	ΔI_{SCD} (load)	14	V_{IN} = 10 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = 10 V		-2.00	-0.01	A	1/ 4/
		16	V_{IN} = 25 V		-1.50	-0.01	"	
		17	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	18	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	19	V_{IN} = 6 V	I_L = -5 mA	4.75	5.25	V	
		20		I_L = -500 mA	"	"	"	
		21	V_{IN} = 20 V	I_L = -5 mA	"	"	"	
		22		I_L = -500 mA	"	"	"	
		23	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		24		I_L = -50 mA	"	"	"	
	V_{RLINE}	25	8 V ≤ V_{IN} ≤ 35 V	I_L = -50 mA	-150	+150	mV	See figure 8 waveforms
		26	8 V ≤ V_{IN} ≤ 25 V	I_L = -350 mA	-50	+50	"	
	V_{RLOAD}	27	V_{IN} = 10 V	-500 mA ≤ I_L ≤ -5 mA	-100	+100	"	
		28	V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA	-150	+150	"	
	I_{SCD}	29	V_{IN} = 10 V	I_L = -5 mA	-7.0	-0.5	mA	
		30	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	31	8 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{30} - I_{31}$
	ΔI_{SCD} (load)	32	V_{IN} = 10 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{29} - I_{32}$
	I_{OS}	33	V_{IN} = 10 V		-2.00	-0.01	A	1/ 4/
		34	V_{IN} = 25 V		-1.50	-0.01	"	
		35	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	36	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		-2.0	-0.5	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	37	V_{IN} = 10 V	I_L = -5 mA	4.70	5.30	V	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	38	V_{IN} = 8 V	I_L = -5 mA	4.75	5.25	V	
		39		I_L = -500 mA	"	"	"	
		40	V_{IN} = 20 V	I_L = -5 mA	"	"	"	
		41		I_L = -500 mA	"	"	"	
		42	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		43		I_L = -50 mA	"	"	"	
	V_{RLINE}	44	8 V $\leq V_{IN} \leq$ 35 V	I_L = -50 mA	-150	+150	mV	See figure 8 waveforms
		45	8 V $\leq V_{IN} \leq$ 25 V	I_L = -350 mA	-50	+50	"	
	V_{RLOAD}	46	V_{IN} = 10 V	-500 mA $\leq I_L \leq$ -5 mA	-100	+100	"	
		47	V_{IN} = 35 V	-50 mA $\leq I_L \leq$ -5 mA	-150	+150	"	
	I_{SCD}	48	V_{IN} = 10 V	I_L = -5 mA	-7.0	-0.5	mA	
		49	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	50	8 V $\leq V_{IN} \leq$ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{49} - I_{50}$
	ΔI_{SCD} (load)	51	V_{IN} = 10 V	-500 mA $\leq I_L \leq$ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{48} - I_{51}$
	I_{OS}	52	V_{IN} = 10 V		-2.00	-0.01	A	1/ 4/
		53	V_{IN} = 25 V		-1.50	-0.01	"	
		54	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	55	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	56	V_{IN} = 10 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	60		dB	See figure 10 2/
7 T_A = +25°C	N_O	57	V_{IN} = 10 V	I_L = -50 mA		125	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	58	V_{IN} = 10 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	59	V_{IN} = 10 V	I_L = -50 mA ΔI_L = -200 mA		2.5	mV/mA	See figure 13
	V_{OUT}	60	V_{IN} = 10 V	I_L = -5 mA	4.75	5.25	V	See figure 8
8 T_A = +125°C	V_{OUT}	61	V_{IN} = 10 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	62	V_{IN} = 10 V	I_L = -5 mA	-2.0	+2.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{61}) \times 10$
8 T_A = -55°C	V_{OUT}	63	V_{IN} = 10 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	64	V_{IN} = 10 V	I_L = -5 mA	-2.0	+2.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{63}) \times 12.5$

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 15 V	I_L = -5 mA	11.40	12.60	V	
		2		I_L = -0.5 A	"	"	"	
		3	V_{IN} = 27 V	I_L = -5 mA	"	"	"	
		4		I_L = -0.5 A	"	"	"	
		5	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		6		I_L = -50 mA	"	"	"	
	V_{RLINE}	7	15 V ≤ V_{IN} ≤ 35 V	I_L = -50 mA	-360	+360	mV	See figure 8 waveforms
		8	15 V ≤ V_{IN} ≤ 32 V	I_L = -350 mA	-120	+120	"	
	V_{RLOAD}	9	V_{IN} = 17 V	-500 mA ≤ I_L ≤ -5 mA	-240	+240	"	
		10	V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA	-360	+360	"	
	I_{SCD}	11	V_{IN} = 17 V	I_L = -5 mA	-7.0	-0.5	mA	
		12	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	13	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	ΔI_{SCD} (load)	14	V_{IN} = 17 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = 17 V		-1.75	-0.01	A	1/ 4/
		16	V_{IN} = 32 V		-1.25	-0.01	"	
		17	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	18	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03 - Continued 3/.

Subgroup	Symbol	Test no.	Conditions (figure 6, unless otherwise specified) Input voltage	Conditions (figure 6, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	19	V_{IN} = 15 V	I_L = -5 mA	11.40	12.60	V	
		20		I_L = -0.5 A	"	"	"	
		21	V_{IN} = 27 V	I_L = -5 mA	"	"	"	
		22		I_L = -0.5 A	"	"	"	
		23	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		24		I_L = -50 mA	"	"	"	
	V_{RLINE}	25	15 V ≤ V_{IN} ≤ 35 V	I_L = -50 mA	-360	+360	mV	See figure 8 waveforms
		26	15 V ≤ V_{IN} ≤ 32 V	I_L = -350 mA	-120	+120	"	
	V_{RLOAD}	27	V_{IN} = 17 V	-500 mA ≤ I_L ≤ -5 mA	-240	+240	"	
		28	V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA	-360	+360	"	
	I_{SCD}	29	V_{IN} = 17 V	I_L = -5 mA	-7.0	-0.5	mA	
		30	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	31	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{30} - I_{31}$
	ΔI_{SCD} (load)	32	V_{IN} = 17 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{29} - I_{32}$
	I_{OS}	33	V_{IN} = 17 V		-1.75	-0.01	A	1/ 4/
		34	V_{IN} = 32 V		-1.25	-0.01	"	
		35	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	36	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		-2.0	-0.5	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	37	V_{IN} = 17 V	I_L = -5 mA	11.28	12.72	V	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	38 39	V_{IN} = 15 V	I_L = -5 mA I_L = -0.5 A	11.40 "	12.60 "	V "	
		40 41	V_{IN} = 27 V	I_L = -5 mA I_L = -0.5 A	" "	" "	" "	
		42 43	V_{IN} = 35 V	I_L = -5 mA I_L = -50 mA	" "	" "	" "	
	V_{RLINE}	44 45	15 V ≤ V_{IN} ≤ 35 V 15 V ≤ V_{IN} ≤ 32 V	I_L = -50 mA I_L = -350 mA	-360 -120	+360 +120	mV "	See figure 8 waveforms
	V_{RLOAD}	46 47	V_{IN} = 17 V V_{IN} = 35 V	-500 mA ≤ I_L ≤ -5 mA -50 mA ≤ I_L ≤ -5 mA	-240 -360	+240 +360	" "	
	I_{SCD}	48 49	V_{IN} = 17 V V_{IN} = 35 V	I_L = -5 mA I_L = -5 mA	-7.0 -8.0	-0.5 -0.5	mA "	
	ΔI_{SCD} (line)	50	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{49} - I_{50}$
	ΔI_{SCD} (load)	51	V_{IN} = 17 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{48} - I_{51}$
	I_{OS}	52 53 54	V_{IN} = 17 V V_{IN} = 32 V V_{IN} = 35 V		-1.75 -1.25 -1.00	-0.01 -0.01 -0.01	A " "	1/ 4/
	I_{OL}	55	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 6, unless otherwise specified) Input voltage	Conditions (figure 6, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	56	V_{IN} = 17 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	55		dB	See figure 10 2/
7 T_A = +25°C	N_O	57	V_{IN} = 17 V	I_L = -50 mA		250	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	58	V_{IN} = 17 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	59	V_{IN} = 17 V	I_L = -50 mA ΔI_L = -200 mA		2.5	mV/mA	See figure 13
	V_{OUT}	60	V_{IN} = 17 V	I_L = -5 mA	11.40	12.60	V	See figure 8
8 T_A = +125°C	V_{OUT}	61	V_{IN} = 17 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	62	V_{IN} = 17 V	I_L = -5 mA	-3.0	+3.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{61}) \times 10$
8 T_A = -55°C	V_{OUT}	63	V_{IN} = 17 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	64	V_{IN} = 17 V	I_L = -5 mA	-3.0	+3.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{63}) \times 12.5$

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 18.5 V	I_L = -5 mA	14.25	15.75	V	
		2		I_L = -500 mA	"	"	"	
		3	V_{IN} = 30 V	I_L = -5 mA	"	"	"	
		4		I_L = -500 mA	"	"	"	
		5	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		6		I_L = -50 mA	"	"	"	
	V_{RLINE}	7	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -350 mA	-150	+150	mV	See figure 8 waveforms
	V_{RLOAD}	8	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA	-300	+300	"	
		9	V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA	-450	+450	"	
	I_{SCD}	10	V_{IN} = 20 V	I_L = -5 mA	-7.0	-0.5	mA	
		11	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	12	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{11} - I_{12}$
	ΔI_{SCD} (load)	13	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{10} - I_{13}$
	I_{OS}	14	V_{IN} = 20 V		-1.75	-0.01	A	1/ 4/
		15	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	16	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 6, unless otherwise specified) Input voltage	Conditions (figure 6, unless otherwise specified) Load current	Limits min	Limits max	Units	Notes
2 T_A = +125°C	V_{OUT}	17	V_{IN} = 16.5 V	I_L = -5 mA	14.25	15.75	V	
		18		I_L = -500 mA	"	"	"	
		19	V_{IN} = 30 V	I_L = -5 mA	"	"	"	
		20		I_L = -500 mA	"	"	"	
		21	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		22		I_L = -50 mA	"	"	"	
	V_{RLINE}	23	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -350 mA	-150	+150	mV	See figure 8 waveforms
	V_{RLOAD}	24	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA	-300	+300	"	
		25	V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA	-450	+450	"	
	I_{SCD}	26	V_{IN} = 20 V	I_L = -5 mA	-7.0	-0.5	mA	
		27	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	28	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{27} - I_{28}$
	ΔI_{SCD} (load)	29	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{26} - I_{29}$
	I_{OS}	30	V_{IN} = 20 V		-1.75	-0.01	A	1/ 4/
		31	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	32	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		-2.0	-0.5	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	33	V_{IN} = 20 V	I_L = -5 mA	14.10	15.90	V	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 6, unless otherwise specified) Input voltage	Conditions (figure 6, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	34	V_{IN} = 18.5 V	I_L = -5 mA	14.25	15.75	V	
		35		I_L = -500 mA	"	"	"	
		36	V_{IN} = 30 V	I_L = -5 mA	"	"	"	
		37		I_L = -500 mA	"	"	"	
		38	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		39		I_L = -50 mA	"	"	"	
	V_{RLINE}	40	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -350 mA	-150	+150	mV	See figure 6 waveforms
	V_{RLOAD}	41	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA	-300	+300	"	
		42	V_{IN} = 35 V	-50 mA ≤ I_L ≤ -5 mA	-450	+450	"	
	I_{SCD}	43	V_{IN} = 20 V	I_L = -5 mA	-7.0	-0.5	mA	
		44	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	45	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{44} - I_{45}$
	ΔI_{SCD} (load)	46	V_{IN} = 20 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{43} - I_{46}$
	I_{OS}	47	V_{IN} = 20 V		-1.75	-0.01	A	1/ 4/
		48	V_{IN} = 35 V		-1.00	-0.01	"	
	I_{OL}	49	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	50	V_{IN} = 20 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	53		dB	See figure 10 2/
7 T_A = +25°C	N_o	51	V_{IN} = 20 V	I_L = -50 mA		300	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	52	V_{IN} = 20 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	53	V_{IN} = 20 V	I_L = -50 mA ΔI_L = -200 mA		2.5	mV/mA	See figure 13
	V_{OUT}	54	V_{IN} = 20 V	I_L = -5 mA	14.25	15.75	V	See figure 8
8 T_A = +125°C	V_{OUT}	55	V_{IN} = 20 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	56	V_{IN} = 20 V	I_L = -5 mA	-3.75	+3.75	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{54} - V_{55}) \times 10$
8 T_A = -55°C	V_{OUT}	57	V_{IN} = 20 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	58	V_{IN} = 20 V	I_L = -5 mA	-3.75	+3.75	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{54} - V_{57}) \times 12.5$

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 28 V	I_L = -5 mA	22.80	25.20	V	
		2		I_L = -500 mA	"	"	"	
		3	V_{IN} = 38 V	I_L = -5 mA	"	"	"	
		4		I_L = -500 mA	"	"	"	
		5	V_{IN} = 40 V	I_L = -5 mA	"	"	"	
		6		I_L = -50 mA	"	"	"	
	V_{RLINE}	7	28 V ≤ V_{IN} ≤ 40 V	I_L = -50 mA	-720	+720	mV	See figure 8 waveforms
		8	28 V ≤ V_{IN} ≤ 38 V	I_L = -350 mA	-240	+240	"	
	V_{RLOAD}	9	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA	-480	+480	"	
		10	V_{IN} = 40 V	-50 mA ≤ I_L ≤ -5 mA	-720	+720	"	
	I_{SCD}	11	V_{IN} = 30 V	I_L = -5 mA	-7.0	-0.5	mA	
		12	V_{IN} = 40 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	13	28 V ≤ V_{IN} ≤ 40 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	ΔI_{SCD} (load)	14	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = 30 V		-1.25	-0.01	A	1/ 4/
		16	V_{IN} = 38 V		-1.00	-0.01	"	
		17	V_{IN} = 40 V		-1.00	-0.01	"	
	I_{OL}	18	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	19	V_{IN} = 28 V	I_L = -5 mA	22.80	25.20	V	
		20		I_L = -500 mA	"	"	"	
		21	V_{IN} = 38 V	I_L = -5 mA	"	"	"	
		22		I_L = -500 mA	"	"	"	
		23	V_{IN} = 40 V	I_L = -5 mA	"	"	"	
		24		I_L = -50 mA	"	"	"	
	V_{RLINE}	25	28 V ≤ V_{IN} ≤ 40 V	I_L = -50 mA	-720	+720	mV	See figure 8 waveforms
		26	28 V ≤ V_{IN} ≤ 38 V	I_L = -350 mA	-240	+240	"	
	V_{RLOAD}	27	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA	-480	+480	"	
		28	V_{IN} = 40 V	-50 mA ≤ I_L ≤ -5 mA	-720	+720	"	
	I_{SCD}	29	V_{IN} = 30 V	I_L = -5 mA	-7.0	-0.5	mA	
		30	V_{IN} = 40 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	31	28 V ≤ V_{IN} ≤ 40 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{30} - I_{31}$
	ΔI_{SCD} (load)	32	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{29} - I_{32}$
	I_{OS}	33	V_{IN} = 30 V		-1.25	-0.01	A	1/ 4/
		34	V_{IN} = 38 V		-1.00	-0.01	"	
		35	V_{IN} = 40 V		-1.00	-0.01	"	
	I_{OL}	36	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		-2.0	-0.5	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	37	V_{IN} = 30 V	I_L = -5 mA	22.56	25.44	V	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	38	V_{IN} = 28 V	I_L = -5 mA	22.80	25.20	V	
		39		I_L = -500 mA	"	"	"	
		40	V_{IN} = 38 V	I_L = -5 mA	"	"	"	
		41		I_L = -500 mA	"	"	"	
		42	V_{IN} = 40 V	I_L = -5 mA	"	"	"	
		43		I_L = -50 mA	"	"	"	
	V_{RLINE}	44	28 V ≤ V_{IN} ≤ 40 V	I_L = -50 mA	-720	+720	mV	See figure 8 waveforms
		45	28 V ≤ V_{IN} ≤ 38 V	I_L = -350 mA	-240	+240	"	
	V_{RLOAD}	46	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA	-480	+480	"	
		47	V_{IN} = 40 V	-50 mA ≤ I_L ≤ -5 mA	-720	+720	"	
	I_{SCD}	48	V_{IN} = 30 V	I_L = -5 mA	-7.0	-0.5	mA	
		49	V_{IN} = 40 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	50	28 V ≤ V_{IN} ≤ 40 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{49} - I_{50}$
	ΔI_{SCD} (load)	51	V_{IN} = 30 V	-500 mA ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{48} - I_{51}$
	I_{OS}	52	V_{IN} = 30 V		-1.25	-0.01	A	1/ 4/
		53	V_{IN} = 38 V		-1.00	-0.01	"	
		54	V_{IN} = 40 V		-1.00	-0.01	"	
	I_{OL}	55	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		-2.0	-0.5	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	56	V_{IN} = 30 V e_i = 1 V rms at f = 2,400 Hz	I_L = -125 mA	50		dB	See figure 10 2/
7 T_A = +25°C	N_o	57	V_{IN} = 30 V	I_L = -50 mA		500	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	58	V_{IN} = 30 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	59	V_{IN} = 30 V	I_L = -50 mA ΔI_L = -200 mA		2.5	mV/mA	See figure 13
	V_{OUT}	60	V_{IN} = 30 V	I_L = -5 mA	22.80	25.20	V	See figure 8
8 T_A = +125°C	V_{OUT}	61	V_{IN} = 30 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	62	V_{IN} = 30 V	I_L = -5 mA	-6.0	+6.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{61}) \times 10$
8 T_A = -55°C	V_{OUT}	63	V_{IN} = 30 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	64	V_{IN} = 30 V	I_L = -5 mA	-6.0	+6.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{63}) \times 12.5$

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TABLE III. Group A inspection for device type 06. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 8 V	I_L = -5 mA	4.75	5.25	V	
		2		I_L = -1.0 A	"	"	"	
		3	V_{IN} = 20 V	I_L = -5 mA	"	"	"	
		4		I_L = -1.0 A	"	"	"	
		5	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		6		I_L = -0.1 A	"	"	"	
	V_{RLINE}	7	8 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A	-150	+150	mV	See figure 8 waveforms
		8	8 V ≤ V_{IN} ≤ 25 V	I_L = -0.5 A	-50	+50	"	
	V_{RLOAD}	9	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA	-100	+100	"	
		10	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-150	+150	"	
	I_{SCD}	11	V_{IN} = 10 V	I_L = -5 mA	-7.0	-0.5	mA	
		12	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	13	8 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	ΔI_{SCD} (load)	14	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = 10 V		-4.00	-0.02	A	1/ 4/
		16	V_{IN} = 25 V		-3.00	-0.02	"	
		17	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	18	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		-4.0	-1.0	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 06 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	19	V_{IN} = 8 V	I_L = -5 mA	4.75	5.25	V	
		20		I_L = -1.0 A	"	"	"	
		21	V_{IN} = 20 V	I_L = -5 mA	"	"	"	
		22		I_L = -1.0 A	"	"	"	
		23	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		24		I_L = -0.1 A	"	"	"	
	V_{RLINE}	25	8 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A	-150	+150	mV	See figure 8 waveforms
		26	8 V ≤ V_{IN} ≤ 25 V	I_L = -0.5 A	-50	+50	"	
	V_{RLOAD}	27	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA	-100	+100	"	
		28	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-150	+150	"	
	I_{SCD}	29	V_{IN} = 10 V	I_L = -5 mA	-7.0	-0.5	mA	
		30	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	31	8 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{30} - I_{31}$
	ΔI_{SCD} (load)	32	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{29} - I_{32}$
	I_{OS}	33	V_{IN} = 10 V		-4.00	-0.02	A	1/ 4/
		34	V_{IN} = 25 V		-3.00	-0.02	"	
		35	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	36	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		-4.0	-1.0	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	37	V_{IN} = 10 V	I_L = -5 mA	4.70	5.30	V	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 06 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	38	V_{IN} = 8 V	I_L = -5 mA	4.75	5.25	V	
		39		I_L = -1.0 A	"	"	"	
		40	V_{IN} = 20 V	I_L = -5 mA	"	"	"	
		41		I_L = -1.0 A	"	"	"	
		42	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		43		I_L = -0.1 A	"	"	"	
	V_{RLINE}	44	8 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A	-150	+150	mV	See figure 8 waveforms
		45	8 V ≤ V_{IN} ≤ 25 V	I_L = -0.5 A	-50	+50	"	
	V_{RLOAD}	46	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA	-100	+100	"	
		47	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-150	+150	"	
	I_{SCD}	48	V_{IN} = 10 V	I_L = -5 mA	-7.0	-0.5	mA	
		49	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	50	8 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{49} - I_{50}$
	ΔI_{SCD} (load)	51	V_{IN} = 10 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{48} - I_{51}$
	I_{OS}	52	V_{IN} = 10 V		-4.00	-0.02	A	1/ 4/
		53	V_{IN} = 25 V		-3.00	-0.02	"	
		54	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	55	V_{IN} = 8 V, forced ΔV_{OUT} = -0.48 V		-4.0	-1.0	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 06 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	56	V_{IN} = 10 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	60		dB	See figure 10 2/
7 T_A = +25°C	N_O	57	V_{IN} = 10 V	I_L = -0.1 A		125	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	58	V_{IN} = 10 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	59	V_{IN} = 10 V	I_L = -100 mA ΔI_L = -400 mA		2.5	mV/mA	See figure 13
	V_{OUT}	60	V_{IN} = 10 V	I_L = -5 mA	4.75	5.25	V	See figure 8
8 T_A = +125°C	V_{OUT}	61	V_{IN} = 10 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	62	V_{IN} = 10 V	I_L = -5 mA	-2.0	+2.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{61}) \times 10$
8 T_A = -55°C	V_{OUT}	63	V_{IN} = 10 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	64	V_{IN} = 10 V	I_L = -5 mA	-2.0	+2.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{63}) \times 12.5$

See footnotes at end of table.

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TABLE III. Group A inspection for device type 07. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 15 V	I_L = -5 mA	11.40	12.60	V	
		2		I_L = -1.0 A	"	"	"	
		3	V_{IN} = 27 V	I_L = -5 mA	"	"	"	
		4		I_L = -1.0 A	"	"	"	
		5	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		6		I_L = -0.1 A	"	"	"	
	V_{RLINE}	7	15 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A	-360	+360	mV	See figure 8 waveforms
		8	15 V ≤ V_{IN} ≤ 32 V	I_L = -0.5 A	-120	+120	"	
	V_{RLOAD}	9	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA	-240	+240	"	
		10	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-360	+360	"	
	I_{SCD}	11	V_{IN} = 17 V	I_L = -5 mA	-7.0	-0.5	mA	
		12	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	13	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	ΔI_{SCD} (load)	14	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = 17 V		-3.50	-0.02	A	1/ 4/
		16	V_{IN} = 32 V		-2.50	-0.02	"	
		17	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	18	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		-4.0	-1.0	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 07 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	19	V_{IN} = 15 V	I_L = -5 mA	11.40	12.60	V	
		20		I_L = -1.0 A	"	"	"	
		21	V_{IN} = 27 V	I_L = -5 mA	"	"	"	
		22		I_L = -1.0 A	"	"	"	
		23	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		24		I_L = -0.1 A	"	"	"	
	V_{RLINE}	25	15 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A	-360	+360	mV	See figure 8 waveforms
		26	15 V ≤ V_{IN} ≤ 25 V	I_L = -0.5 A	-120	-120	"	
	V_{RLOAD}	27	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA	-240	+240	"	
		28	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-360	+360	"	
	I_{SCD}	29	V_{IN} = 17 V	I_L = -5 mA	-7.0	-0.5	mA	
		30	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	31	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{30} - I_{31}$
	ΔI_{SCD} (load)	32	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{29} - I_{32}$
	I_{OS}	33	V_{IN} = 17 V		-3.50	-0.02	A	1/ 4/
		34	V_{IN} = 32 V		-2.50	-0.02	"	
		35	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	36	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		-4.0	-1.0	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	37	V_{IN} = 17 V	I_L = -5 mA	11.28	12.72	V	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 07 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	38	V_{IN} = 15 V	I_L = -5 mA	11.40	12.60	V	
		39		I_L = -1.0 A	"	"	"	
		40	V_{IN} = 27 V	I_L = -5 mA	"	"	"	
		41		I_L = -1.0 A	"	"	"	
		42	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		43		I_L = -0.1 A	"	"	"	
	V_{RLINE}	44	15 V ≤ V_{IN} ≤ 35 V	I_L = -0.1 A	-360	+360	mV	See figure 8 waveforms
		45	15 V ≤ V_{IN} ≤ 32 V	I_L = -0.5 A	-120	+120	"	
	V_{RLOAD}	46	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA	-240	+240	"	
		47	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-360	+360	"	
	I_{SCD}	48	V_{IN} = 17 V	I_L = -5 mA	-7.0	-0.5	mA	
		49	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	50	15 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{49} - I_{50}$
	ΔI_{SCD} (load)	51	V_{IN} = 17 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{48} - I_{51}$
	I_{OS}	52	V_{IN} = 17 V		-3.50	-0.02	A	1/ 4/
		53	V_{IN} = 32 V		-2.50	-0.02	"	
		54	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	55	V_{IN} = 15 V, forced ΔV_{OUT} = -1.13 V		-4.0	-1.0	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 07 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified)		Limits		Units	Notes
			Input voltage	Load current	Min	Max		
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	56	V_{IN} = 17 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	55		dB	See figure 10 2/
7 T_A = +25°C	N_o	57	V_{IN} = 17 V	I_L = -0.1 A		250	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	58	V_{IN} = 17 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	59	V_{IN} = 17 V	I_L = -100 mA ΔI_L = -400 mA		2.5	mV/mA	See figure 13
	V_{OUT}	60	V_{IN} = 17 V	I_L = -5 mA	11.40	12.60	V	See figure 8
8 T_A = +125°C	V_{OUT}	61	V_{IN} = 17 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	62	V_{IN} = 17 V	I_L = -5 mA	-3.0	+3.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{61}) \times 10$
8 T_A = -55°C	V_{OUT}	63	V_{IN} = 17 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	64	V_{IN} = 17 V	I_L = -5 mA	-3.0	+3.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{60} - V_{63}) \times 12.5$

See footnotes at end of table.

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TABLE III. Group A inspection for device type 08. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 18.5 V	I_L = -5 mA	14.25	15.75	V	
		2		I_L = -1.0 A	"	"	"	
		3	V_{IN} = 30 V	I_L = -5 mA	"	"	"	
		4		I_L = -1.0 A	"	"	"	
		5	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		6		I_L = -0.1 A	"	"	"	
	V_{RLINE}	7	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -0.5 A	-150	+150	mV	See figure 8 waveforms
	V_{RLOAD}	8	V_{IN} = 20 V	-1.0 A ≤ I_L ≤ -5 mA	-300	+300	"	
		9	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-450	+450	"	
	I_{SCD}	10	V_{IN} = 20 V	I_L = -5 mA	-7.0	-0.5	mA	
		11	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	12	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{11} - I_{12}$
	ΔI_{SCD} (load)	13	V_{IN} = 20 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{10} - I_{13}$
	I_{OS}	14	V_{IN} = 20 V		-3.50	-0.02	A	1/ 4/
		15	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	16	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		-4.0	-1.0	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 08 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	17	V_{IN} = 18.5 V	I_L = -5 mA	14.25	15.75	V	
		18		I_L = -1.0 A	"	"	"	
		19	V_{IN} = 30 V	I_L = -5 mA	"	"	"	
		20		I_L = -1.0 A	"	"	"	
		21	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		22		I_L = -0.1 A	"	"	"	
	V_{RLINE}	23	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -0.5 A	-150	+150	mV	See figure 8 waveforms
	V_{RLOAD}	24	V_{IN} = 20 V	-1.0 A ≤ I_L ≤ -5 mA	-300	+300	"	
		25	V_{IN} = 35 V	-0.1 A ≤ I_L ≤ -5 mA	-450	+450	"	
	I_{SCD}	26	V_{IN} = 20 V	I_L = -5 mA	-7.0	-0.5	mA	
		27	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	28	18.5 V ≤ V_{IN} ≤ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{27} - I_{28}$
	ΔI_{SCD} (load)	29	V_{IN} = 20 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{26} - I_{29}$
	I_{OS}	30	V_{IN} = 20 V		-3.50	-0.02	A	1/ 4/
		31	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	32	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		-4.0	-1.0	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	33	V_{IN} = 20 V	I_L = -5 mA	14.10	15.90	V	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 08 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	34	V_{IN} = 18.5 V	I_L = -5 mA	14.25	15.75	V	
		35		I_L = -1.0 A	"	"	"	
		36	V_{IN} = 30 V	I_L = -5 mA	"	"	"	
		37		I_L = -1.0 A	"	"	"	
		38	V_{IN} = 35 V	I_L = -5 mA	"	"	"	
		39		I_L = -0.1 A	"	"	"	
	V_{RLINE}	40	18.5 V $\leq V_{IN} \leq$ 35 V	I_L = -0.5 A	-150	+150	mV	See figure 8 waveforms
	V_{RLOAD}	41	V_{IN} = 20 V	-1.0 A $\leq I_L \leq$ -5 mA	-300	+300	"	
		42	V_{IN} = 35 V	-0.1 A $\leq I_L \leq$ -5 mA	-450	+450	"	
	I_{SCD}	43	V_{IN} = 20 V	I_L = -5 mA	-7.0	-0.5	mA	
		44	V_{IN} = 35 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	45	18.5 V $\leq V_{IN} \leq$ 35 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{44} - I_{45}$
	ΔI_{SCD} (load)	46	V_{IN} = 20 V	-1.0 A $\leq I_L \leq$ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{43} - I_{46}$
	I_{OS}	47	V_{IN} = 20 V		-3.50	-0.02	A	1/ 4/
		48	V_{IN} = 35 V		-2.00	-0.02	"	
	I_{OL}	49	V_{IN} = 18.5 V, forced ΔV_{OUT} = -1.43 V		-4.0	-1.0	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 08 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	50	V_{IN} = 20 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	53		dB	See figure 10 2/
7 T_A = +25°C	N_o	51	V_{IN} = 20 V	I_L = -0.1 A		300	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	52	V_{IN} = 20 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	53	V_{IN} = 20 V	I_L = -100 mA ΔI_L = -400 mA		2.5	mV/mA	See figure 13
	V_{OUT}	54	V_{IN} = 20 V	I_L = -5 mA	14.25	15.75	V	See figure 8
8 T_A = +125°C	V_{OUT}	55	V_{IN} = 20 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	56	V_{IN} = 20 V	I_L = -5 mA	-3.75	+3.75	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{54} - V_{55}) \times 10$
8 T_A = -55°C	V_{OUT}	57	V_{IN} = 20 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	58	V_{IN} = 20 V	I_L = -5 mA	-3.75	+3.75	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{54} - V_{57}) \times 12.5$

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TABLE III. Group A inspection for device type 09. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
1 T_A = +25°C	V_{OUT}	1	V_{IN} = 28 V	I_L = -5 mA	22.80	25.20	V	
		2		I_L = -1.0 A	"	"	"	
		3	V_{IN} = 38 V	I_L = -5 mA	"	"	"	
		4		I_L = -1.0 A	"	"	"	
		5	V_{IN} = 40 V	I_L = -5 mA	"	"	"	
		6		I_L = -0.1 A	"	"	"	
	V_{RLINE}	7	28 V ≤ V_{IN} ≤ 40 V	I_L = -0.1 A	-720	+720	mV	See figure 8 waveforms
		8	28 V ≤ V_{IN} ≤ 38 V	I_L = -0.5 A	-240	+240	"	
	V_{RLOAD}	9	V_{IN} = 30 V	-1.0 A ≤ I_L ≤ -5 mA	-480	+480	"	
		10	V_{IN} = 40 V	-0.1 A ≤ I_L ≤ -5 mA	-720	+720	"	
	I_{SCD}	11	V_{IN} = 30 V	I_L = -5 mA	-7.0	-0.5	mA	
		12	V_{IN} = 40 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	13	28 V ≤ V_{IN} ≤ 40 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{12} - I_{13}$
	ΔI_{SCD} (load)	14	V_{IN} = 30 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{11} - I_{14}$
	I_{OS}	15	V_{IN} = 30 V		-2.50	-0.02	A	1/ 4/
		16	V_{IN} = 35 V		-2.00	-0.005	"	
	I_{OL}	17	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		-4.0	-1.0	"	See figure 9 5/

See footnotes at end of table.

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TABLE III. Group A inspection for device type 09 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
2 T_A = +125°C	V_{OUT}	18	V_{IN} = 28 V	I_L = -5 mA	22.80	25.20	V	
		19		I_L = -1.0 A	"	"	"	
		20	V_{IN} = 38 V	I_L = -5 mA	"	"	"	
		21		I_L = -1.0 A	"	"	"	
		22	V_{IN} = 40 V	I_L = -5 mA	"	"	"	
		23		I_L = -0.1 A	"	"	"	
	V_{RLINE}	24	28 V ≤ V_{IN} ≤ 40 V	I_L = -0.1 A	-720	+720	mV	See figure 8 waveforms
		25	28 V ≤ V_{IN} ≤ 38 V	I_L = -0.5 A	-240	+240	"	
	V_{RLOAD}	26	V_{IN} = 30 V	-1.0 A ≤ I_L ≤ -5 mA	-480	+480	"	
		27	V_{IN} = 40 V	-0.1 A ≤ I_L ≤ -5 mA	-720	+720	"	
	I_{SCD}	28	V_{IN} = 30 V	I_L = -5 mA	-7.0	-0.5	mA	
		29	V_{IN} = 40 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	30	28 V ≤ V_{IN} ≤ 40 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{30} - I_{31}$
	ΔI_{SCD} (load)	31	V_{IN} = 30 V	-1.0 A ≤ I_L ≤ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{29} - I_{32}$
	I_{OS}	32	V_{IN} = 30 V		-2.50	-0.02	A	1/ 4/
		33	V_{IN} = 35 V		-2.00	-0.005	"	
	I_{OL}	34	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		-4.0	-1.0	"	See figure 9 5/
2 T_A = +150°C	V_{OUT}	35	V_{IN} = 30 V	I_L = -5 mA	22.56	25.44	V	

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TABLE III. Group A inspection for device type 09 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Conditions (figure 8, unless otherwise specified) Load current	Limits Min	Limits Max	Units	Notes
3 T_A = -55°C	V_{OUT}	36	V_{IN} = 28 V	I_L = -5 mA	22.80	25.20	V	
		37		I_L = -1.0 A	"	"	"	
		38	V_{IN} = 38 V	I_L = -5 mA	"	"	"	
		39		I_L = -1.0 A	"	"	"	
		40	V_{IN} = 40 V	I_L = -5 mA	"	"	"	
		41		I_L = -0.1 A	"	"	"	
	V_{RLINE}	42	28 V $\leq V_{IN} \leq$ 40 V	I_L = -0.1 A	-720	+720	mV	See figure 8 waveforms
		43	28 V $\leq V_{IN} \leq$ 38 V	I_L = -0.5 A	-240	+240	"	
	V_{RLOAD}	44	V_{IN} = 30 V	-1.0 A $\leq I_L \leq$ -5 mA	-480	+480	"	
		45	V_{IN} = 40 V	-0.1 A $\leq I_L \leq$ -5 mA	-720	+720	"	
	I_{SCD}	46	V_{IN} = 30 V	I_L = -5 mA	-7.0	-0.5	mA	
		47	V_{IN} = 40 V	I_L = -5 mA	-8.0	-0.5	"	
	ΔI_{SCD} (line)	48	28 V $\leq V_{IN} \leq$ 40 V	I_L = -5 mA	-1.0	+1.0	"	$\Delta I_{SCD} = I_{49} - I_{50}$
	ΔI_{SCD} (load)	49	V_{IN} = 30 V	-1.0 A $\leq I_L \leq$ -5 mA	-0.5	+0.5	"	$\Delta I_{SCD} = I_{48} - I_{51}$
	I_{OS}	50	V_{IN} = 30 V		-2.50	-0.02	A	1/ 4/
		51	V_{IN} = 35 V		-2.00	-0.005	"	
	I_{OL}	52	V_{IN} = 28 V, forced ΔV_{OUT} = -2.28 V		-4.0	-1.0	"	See figure 9 5/

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TABLE III. Group A inspection for device type 09 - Continued. 3/

Subgroup	Symbol	Test no.	Conditions (figure 8, unless otherwise specified) Input voltage	Load current	Limits Min	Limits Max	Units	Notes
4 T_A = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{OUT}}$	53	V_{IN} = 30 V e_i = 1 V rms at f = 2,400 Hz	I_L = -350 mA	50		dB	See figure 10 2/
7 T_A = +25°C	N_O	54	V_{IN} = 30 V	I_L = -0.1 A		500	µV rms	See figure 11 2/
	$\frac{\Delta V_{OUT}}{\Delta V_{IN}}$	55	V_{IN} = 30 V V_{pulse} = 3.0 V	I_L = -5 mA		30	mV/V	See figure 12
	$\frac{\Delta V_{OUT}}{\Delta I_L}$	56	V_{IN} = 30 V	I_L = -100 mA ΔI_L = -400 mA		2.5	mV/mA	See figure 13
	V_{OUT}	57	V_{IN} = 30 V	I_L = -5 mA	22.80	25.20	V	See figure 8
8 T_A = +125°C	V_{OUT}	58	V_{IN} = 30 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	59	V_{IN} = 30 V	I_L = -5 mA	-6.0	+6.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{57} - V_{58}) \times 10$
8 T_A = -55°C	V_{OUT}	60	V_{IN} = 30 V	I_L = -5 mA			V	See figure 8
	$\frac{\Delta V_{OUT}}{\Delta T}$	61	V_{IN} = 30 V	I_L = -5 mA	-6.0	+6.0	mV/°C	See figure 8 $\frac{\Delta V_{OUT}}{\Delta T} = (V_{57} - V_{60}) \times 12.5$

1/ One second maximum test duration.

2/ The meter for e_i and e_o shall have a minimum bandwidth from 10 Hz to 10 kHz and shall measure true rms voltages.

3/ For device types 02 through 09, all tests performed at T_A = +125°C may, at the manufacturer's option, be performed at T_A = +150°C. Specifications for T_A = +125°C shall then apply at T_A = +150°C.

4/ A 1 µF capacitor may be added to dampen oscillations during the I_{OS} test for device types 02 through 09.

5/ The overload current test checks the device current limiting, and it is not a measure of the output peak current.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Life test for class S shall be performed at +125°C for 1,000 hours on a quantity of 45 devices with 0 failures (in accordance with group C, subgroup 2) as a substitute for subgroup 5 of table IIa. If the sample fails the quantity (accept number), individual devices in the lot may be acceptable if they pass 1,000 hours additional burn-in and the lot PDA does not exceed 5 percent.

c. Constant acceleration for class S (method 2001 of MIL-STD-883). Test condition B shall be used for case Y.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Life tests for class B (method 1005 of MIL-STD-883): Test condition C only using the circuits shown on figure 7. Accelerated test conditions shall not be used.

c. Constant acceleration (method 2001 of MIL-STD-883). Test condition B shall be used for case Y.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Constant acceleration (method 2001 of MIL-STD-883). Test condition B shall be used for case Y.

4.4.5 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given, except the input offset voltage (or differential voltage) are referenced to the external zero reference level of the supply voltage. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within +10°C of their power stable condition prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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TABLE IV. Group C end-point electrical parameters (T_A = +25°C).

Device	Characteristic	Symbol	Delta limits 1/	Limits		Unit
				Min	Max	
01	Line regulation	V_{RLINE}	±10 mV	-50	+50	mV
	Load regulation	V_{RLOAD}	±10 mV	-50	+50	mV
	Standby current drain	I_{SCD}	±1 mA	-10	0.5	mA
	Output voltage	V_{OUT}	±25 mV	4.70	5.40	V
02, 06	Output voltage	V_{OUT}	±25 mV	4.75	5.25	V
	Standby current drain	I_{SCD}	±20%	-7.0	-0.5	mA
03, 07	Output voltage	V_{OUT}	±60 mV	11.40	12.60	V
	Standby current drain	I_{SCD}	±20%	-7.0	-0.5	mA
04, 08	Output voltage	V_{OUT}	±75 mV	14.25	15.75	V
	Standby current drain	I_{SCD}	±20%	-7.0	-0.5	mA
05, 09	Output voltage	V_{OUT}	±120 mV	22.80	25.20	V
	Standby current drain	I_{SCD}	±20%	-7.0	-0.5	mA

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. PIN (see 6.6).

b. Requirements for delivery of one copy of the quality comformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

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f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable. These requirements shall not affect the PIN.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows.

6.3.1 Line regulation. The change in output voltage for a specified change in input voltage (V_{RLINE}) (for device type 01: "The percentage change").

6.3.2 Load regulation. The change in output voltage for a specified change in load current (V_{RLOAD}) (for device type 01: "The percentage change").

6.3.3 Ripple rejection. The ratio of the peak-to-peak input ripple voltage to the peak-to-peak output ripple voltage ($\frac{\Delta V_{OUT}}{\Delta V_{IN}}$).

6.3.4 Average temperature coefficient of output voltage. The percentage change in output voltage for a specified change in ambient temperature ($\frac{\Delta V_{OUT}}{\Delta T}$).

6.3.5 Output noise voltage. The rms output noise voltage with constant load and no input ripple (N_O).

6.3.6 Standby current drain. The supply current drawn by the regulator with no output load or with a 1 kΩ output load.

6.3.7 Input voltage range. The range of supply voltage over which the regulator will operate.

6.3.8 Output voltage range. The range of output voltage over which the regulator will operate.

6.3.9 Transient response. The closed-loop step function response of the regulator under small-signal conditions.

6.3.10 Start-up input voltage. The minimum input voltage required to obtain the specified output voltage (for device type 01 only).

6.4 Logistic support. Lead material and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead formimg shall not affect the PIN.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	LM109
02	LM140H-05, 78M05
03	LM140H-12, 78M12
04	LM140H-15, 78M15
05	LM140H-24, 78M24
06	LM140K-05, 7805
07	LM140K-12, 7812
08	LM140K-15, 7815
09	LM140K-24, 7824

6.6 PIN. The PIN shall be in accordance with MIL-M-38510.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify change with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1145)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/107C	MICROCIRCUITS, LINEAR, POSITIVE, VOLTAGE REGULATORS,

3a. NAME OF SUBMITTING ORGANIZATION MONOLITHIC CILICON	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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